

Data Science Concepts

Lesson05–Unsupervised Learning

Objective

After completing this lesson you will be able to:



- Understand unsupervised learning and its applications
- Describe hierarchical clustering and K means clustering
 - Understand limitations and considerations
- Understand various distance measuring techniques
- Explain Quality of Cluster
- Describe DBSCAN clustering and its limitations
- Understand the working of HDBSCAN
- Describe Gaussian Model for anomaly detection
- Understand Principle component analysis
 - Usage of PCA as a tool for exploration
 - Usage in supervised learning
 - PCA and its use in data imputation
 - PCA as a tool to build recommender system

Unsupervised learning

- A set of **statistical and machine learning** tools intended for settings where the goal is **to discover interesting pattern** from a set of features (Xs) measured on 'n' observations:
 - Can there be informative way to visualize the data?
 - Are there subgroups amongst the variables/observation?
- Challenging compared to supervised learning as there is no ground truth
 - No universally accepted mechanism – Cross validation or validation on independent data set.
- Still useful for exploration
 - Trying to identify subgroups amongst breast cancer people or gene to better understand the disease.
 - Identify subgroups of shoppers based on browsing pattern, purchase history, items purchased etc.

Cluster Analysis

- Used in marketing for creating product segmentation and customer segmentation.
 - Is helpful to understand the product spread and understand which products are cannibalizing. Either internal or of the competitors.
 - Helps in creating customer profiles for targeted marketing.
 - The marketing expense can be optimized and utilized effectively.
- Clustering:
 - Putting similar things into one single group.
 - Clustering is performed by looking into different characteristics which may be helpful in bringing out a pattern.

Types of clustering being discussed:

- Hierarchical clustering
- K means clustering
- DBSCAN
- HDBSCAN
- Gaussian Model

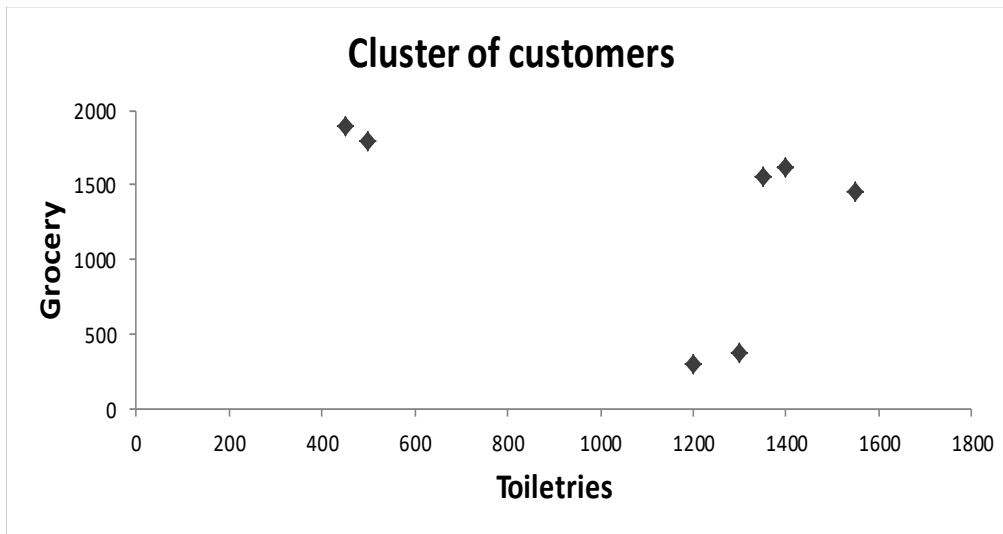
Hierarchical Clustering

Hierarchical Clustering–Concept Development

Customer	Groceries	Toiletries
1	1200	300
2	1300	380
3	500	1800
4	450	1900
5	1350	1560
6	1400	1620
7	1550	1450

Three distinctively different categories:

- Low on toiletries and high on grocery
- High on toiletries and low on grocery
- High on toiletries and high on grocery



How do you do this for 10000 customers and 20 products?

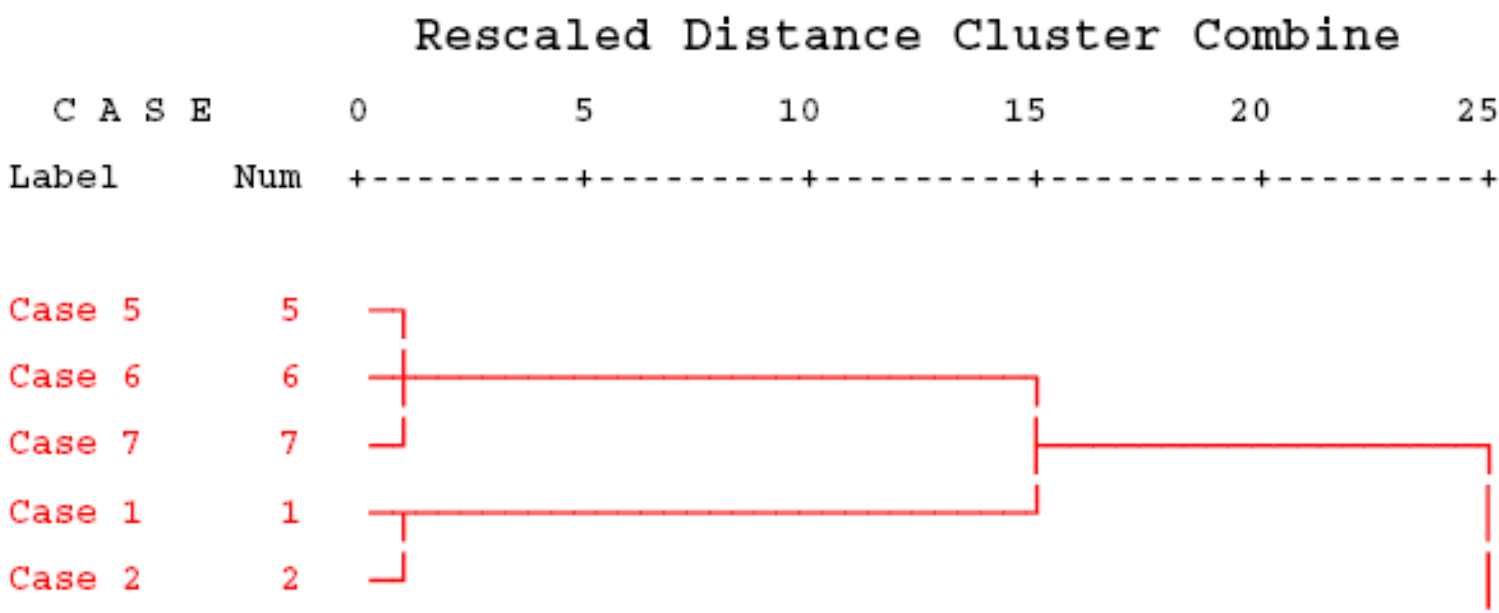
Hierarchical Clustering–Concept Development

- Closer the points were, more similarity within the customers. Farther the points, more dissimilarity within the customers.
 - Distance between the points is a measure of similarity or dissimilarity
- Calculate the linear distance between all the customers:
 - Distance between Customer 1 to Customer 2, 3, ..., 7.
 - Distance between Customer 2 to Customer 1, 3, ..., 7.
 - Distance between Customer 3 to Customer 1, 2, ..., 7.
 - .
 - .
 - Distance between Customer 7 to Customer 1, 2, ..., 6.
- 7*7 matrix is formed. Pick the smallest number. This forms the first cluster.
- Now one cluster and 5 customers. Total six entities for which above steps are repeated. Cluster formation happens in hierarchy and thus the name

Hierarchical Clustering–Concept Development

- When to stop the clustering:
 - The variation within the cluster is low and variation across cluster is very high.
 - Dendrogram gives this output in graphical form.
 - Farther distance travelled on dendrogram, more dissimilar entities are being clustered.

Dendrogram using Average Linkage (Between Groups)



Hierarchical Clustering–Beer Data Case

20 brands of beer with Calorie content, sodium content, Alcohol content and Cost.

ID	BEER	CAL	SOD	ALC	COST
1	Budweiser	144	15	4.7	0.43
2	Schlitz	151	19	4.9	0.43
3	Lowenbrau	157	15	4.9	0.48
4	Kronenbourg	170	7	5.2	0.73
5	Heineken	152	11	5	0.77
6	Old Mil	145	23	4.6	0.28
7	Augsburger	175	24	5.5	0.4
8	Strohs	149	27	4.7	0.42
9	Miller lite	99	10	4.3	0.43
10	Bud light	113	8	3.7	0.44
11	Coors	140	18	4.6	0.44
12	Coors lite	102	15	4.1	0.45
13	Michelob light	135	11	4.2	0.5
14	Becks	150	19	4.7	0.76
15	Kirin	149	6	5	0.79
16	Pabst	68	15	2.3	0.38
17	Hamms	136	19	4.4	0.43
18	Heilemans	144	24	4.9	0.43
19	Olympia	72	6	2.9	0.46
20	Schilitz lite	97	7	4.2	0.47

Perform Hierarchical Clustering

Hierarchical Clustering–Beer Data Case

Euclidian distance from Budweiser to all other beer brands

ID	BEER	CAL	SOD	ALC	COST
1	Budweiser	144	15	4.7	0.43
2	Schlitz	151	19	4.9	0.43
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16	Pabst	68	15	2.3	0.38
17	Hamms	136	19	4.4	0.43
18	Heilemans	144	24	4.9	0.43
19	Olympia	72	6	2.9	0.46
20	Schilitz lite	97	7	4.2	0.47

Original Data

ID	BEER	CAL	SOD	ALC	COST	Total
1	Budweiser	0	0	0	0.000	0.000
2	Schlitz	49	16	0.04	0.000	8.065
3	Lowenbrau	169	0	0.04	0.003	13.002
4	Kronenbourg	676	64	0.25	0.090	27.209
5	Heineken	64	16	0.09	0.116	8.956
6	Old Mil	1	64	0.01	0.023	8.064
7	Augsburger	961	81	0.64	0.001	32.290
8	Strohs	25	144	0	0.000	13.000
9	Miller lite	2025	25	0.16	0.000	45.279
10	Bud light	961	49	1	0.000	31.796
11	Coors	16	9	0.01	0.000	5.001
12	Coors lite	1764	0	0.36	0.000	42.004
13	Michelob light	81	16	0.25	0.005	9.862
14	Becks	36	16	0	0.109	7.219
15	Kirin	25	81	0.09	0.130	10.306
16	Pabst	5776	0	5.76	0.003	76.038
17	Hamms	64	16	0.09	0.000	8.949
18	Heilemans	0	81	0.04	0.000	9.002
19	Olympia	5184	81	3.24	0.001	72.583
20	Schilitz lite	2209	64	0.25	0.002	47.679

Euclidian distance from Budweiser

These distances are to be calculated for each beer brand



The total column is called the Euclidian distance

$$D(X_1, X_2) = \sqrt{(x_{11} - x_{21})^2 + (x_{12} - x_{22})^2 + \dots + (x_{1n} - x_{2n})^2}$$

Hierarchical Clustering–Beer Data Case

Euclidian distance matrix for all the brands

BEER	Budweiser	Schlitz	Lowenbrau	Kronenbou	Heineken	Old Mil	Augsburger	Strohs	Miller lite	Bud light	Coors
Budweiser	0.00	8.06	13.00	27.21	8.96	8.06	32.29	13.00	45.28	31.80	5.00
Schlitz	8.06	0.00	7.21	22.48	8.07	7.22	24.52	8.25	52.78	39.58	11.05
Lowenbrau	13.00	7.21	0.00	15.27	6.41	14.43	20.13	14.42	58.22	44.57	17.27
Kronenbou	27.21	22.48	15.27	0.00	18.44	29.69	17.73	29.01	71.07	57.03	31.96
Heineken	8.96	8.07	6.41	18.44	0.00	13.91	26.43	16.29	53.02	39.14	13.90
Old Mil	8.06	7.22	14.43	29.69	13.91	0.00	30.03	5.66	47.80	35.35	7.07
Augsburger	32.29	24.52	20.13	17.73	26.43	30.03	0.00	26.18	77.29	64.06	35.52
Strohs	13.00	8.25	14.42	29.01	16.29	5.66	26.18	0.00	52.81	40.72	12.73
Miller lite	45.28	52.78	58.22	71.07	53.02	47.80	77.29	52.81	0.00	14.15	41.77
Bud light	31.80	39.58	44.57	57.03	39.14	35.35	64.06	40.72	14.15	0.00	28.81
Coors	5.00	11.05	17.27	31.96	13.90	7.07	35.52	12.73	41.77	28.81	0.00
Coors lite	42.00	49.17	55.01	68.48	50.17	43.74	73.57	48.51	5.83	13.04	38.12
Michelob li	9.86	17.90	22.37	35.24	17.02	15.63	42.08	21.27	36.01	22.21	8.61
Becks	7.22	1.07	8.07	23.33	8.25	6.42	25.51	8.07	51.79	38.61	10.06
Kirin	10.31	13.16	12.05	21.02	5.83	17.48	31.63	21.01	50.17	36.08	15.01
Pabst	76.04	83.14	89.04	102.35	84.14	77.45	107.43	81.92	31.46	45.56	72.10
Hamms	8.95	15.01	21.38	36.07	17.90	9.85	39.33	15.27	38.08	25.50	4.13
Heilemans	9.00	8.60	15.81	31.07	15.27	1.45	31.01	5.83	47.13	34.91	7.22
Olympia	72.58	80.09	85.50	98.03	80.18	74.97	104.59	79.83	27.33	41.06	69.07
Schilitz lite	47.68	8.00	0.50	0.00	48.35	50.60	79.84	55.72	3.61	16.04	44.39

But there may be a problem if clustering is done without standardizing the data.

- High frequency purchases or unit of measurement has impact on inter entity dissimilarity.

Amalgamation or Linkage Rules

Once several objects have been linked together, how do we determine the distances between those new clusters?

Single linkage (nearest neighbor):

- The distance between two clusters is determined by the distance of the two closest objects (nearest neighbors) in the different clusters.
- Group cluster in a bottom up (agglomerative) fashion such that at each step it combines closest pair of elements not belonging to the same cluster.

$$D(X, Y) = \min_{x \in X, y \in Y} d(x, y)$$

Complete linkage (furthest neighbor):

- The distances between clusters are determined by the greatest distance between any two objects in the different clusters (i.e., by the "furthest neighbors").

$$D(X, Y) = \max_{x \in X, y \in Y} d(x, y)$$

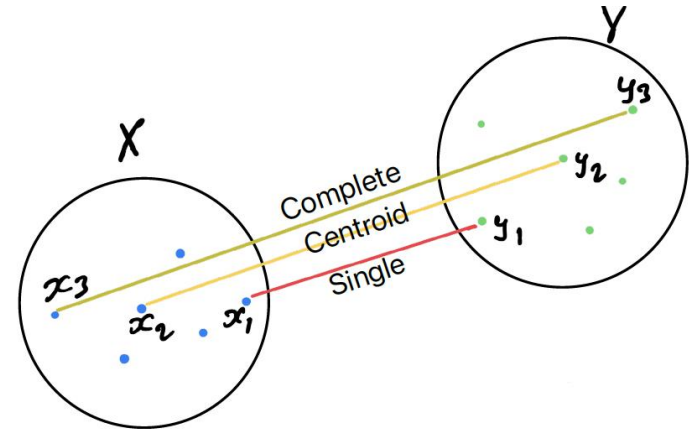


X and Y are any two cluster. $d(x, y)$ is the distance between two points x and y where $x \in X$ and $y \in Y$.

Useful link: <https://online.stat.psu.edu/stat555/node/86/> and <https://www.analyticsvidhya.com/blog/2019/05/beginners-guide-hierarchical-clustering/>

Linkage strategy and limitation

- Single or complete linkage is highly influenced by noise.
 - a single noise data point in the wrong place can act as a bridge between clusters and can glue them together.
- Main problem with hierarchical clustering algorithms - deriving appropriate parameters for the termination condition.
 - The value at which to cut-off a cluster should be small enough to separate all “natural” clusters and, at the same time large enough such that no cluster is split into two parts.



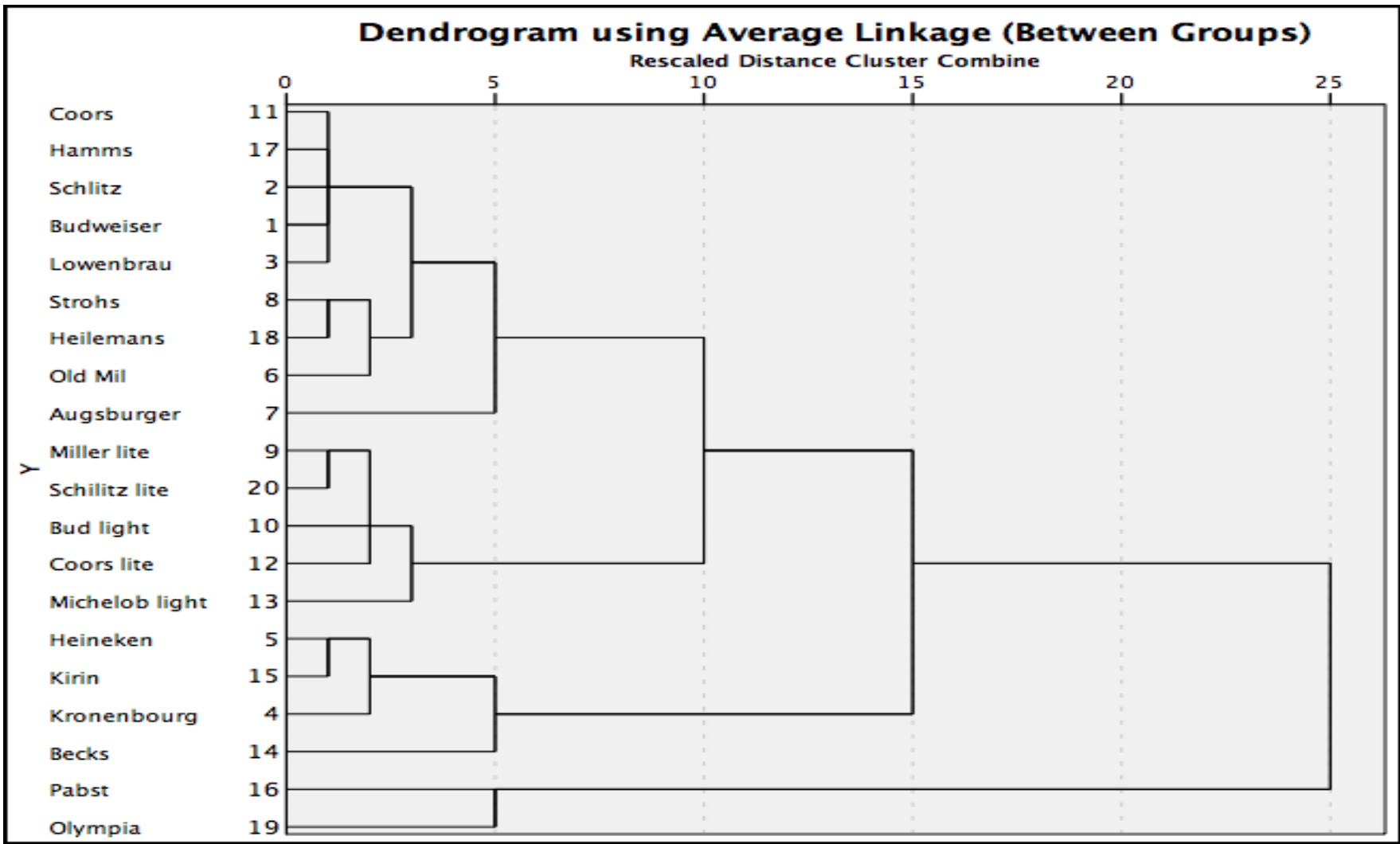
Say, “1 and 3” is a part of one cluster and 2 is a separate element then:

- Single Linkage:
$$D(13, 2) = \text{Min}[d(1,2), d(3,2)]$$
- Complete Linkage:
$$D(13, 2) = \text{Max}[d(1,2), d(3,2)]$$



https://hdbscan.readthedocs.io/en/latest/comparing_clustering_algorithms.html#agglomerative-clustering. Ward's linkage minimizes the inc. in variance when two clusters are merged.

Hierarchical Clustering–Beer Data Case



Interpreting Dendrogram

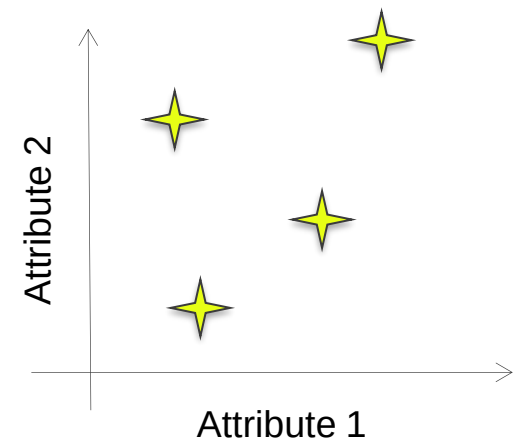
- Leaf represent the 20 beer brands. They fuse together as we move from left to right.
 - The fused branches, fuse with other leaf or branches in similar fashion.
- The earlier the fusion, more similar the group of beer brands (observations).
 - The length of the fusion (distance) indicates the difference between observations.
- Any number of clusters can be obtained: based on distance of the fusion and the number of clusters desired.

K Means Clustering

K Means Clustering–Concept Development

- Assume 70, 000 customer data having two attributes which needs to be segmented in 4 clusters.
- Here 4 depicts the K value for the cluster to be formed. Steps in K – means clustering
 1. Map 70K data points to 4 random clusters centroids. Called cluster assignment step.
 2. Once 70K numbers are mapped, calculate the new centroid. There will be four centroids. This is centroid movement step.
 3. Remove the data points and keep the new centroids.
- Repeat step 2 to 3 till the centroid movement stops.

Output is 4 clusters mapping 70, 000 customer data.



Read the difference K-means and K-medoids: <https://online.stat.psu.edu/stat555/node/88/>

K Means Clustering (Algorithm approach)

K- Means

- Initial cluster assignment: Randomly assign a number from 1 to K to each of the observations.
- Iterate until cluster assignment stops changing:
 - For each of the K clusters, compute cluster centroid. This is assignment step.
 - Assign each observation to cluster whose centroid is closet based on Euclidian distance.
- Good cluster is one where within cluster variation $W(C_k)$ is the smallest; C_k is the cluster indices containing finite set of observations.

K Means Algorithm

Randomly initialize K Cluster Centroids ($\mu_1, \mu_2, \mu_3 \dots \mu_k$)

Training set is $\{x^{(1)}, x^{(2)}, x^{(3)}, \dots x^{(m)}\}$

Cluster assignment step

Repeat { $i = 1$ to m
 $c^{(i)}$ = index from 1 to K of cluster centroid closet to $x^{(i)}$
}

Move Centroid

Repeat { $k = 1$ to K
 μ_k = average of points assigned to cluster k
}

Cost function for K Means

$c^{(i)}$ = index of cluster (1, 2, ... K) to which $x^{(i)}$ is currently assigned

μ_k = cluster centroid k where $k = \{1, 2, 3, \dots, K\}$

$\mu_{c^{(i)}}$ = cluster centroid of the cluster to which example $x^{(i)}$ has been assigned

Optimization objective

$$J(c^{(1)}, \dots, c^{(m)}, \mu_1, \dots, \mu_k) = 1/m \sum_{i=1}^m \|x^{(i)} - \mu_{c^{(i)}}\|^2$$

$$\underset{\substack{c^{(1)}, \dots, c^{(m)} \\ \mu_1, \dots, \mu_k}}{\text{minimize}} J(c^{(1)}, \dots, c^{(m)}, \mu_1, \dots, \mu_k)$$

Random initialization of K clusters

Pick k random training examples and set $\mu_1, \dots, \mu_k$ equal to these k training examples

```
for{ i = 1 to 100
```

```
    Randomly initialize K means
```

```
    Run K means to get cluster assignment and centroid movement
```

```
    Compute cost function
```

```
}
```

Pick the cluster that gives the lowest cost $J(c^{(1)}, \dots, c^{(m)}, \mu_1, \dots, \mu_k)$

K Nearest Neighbor for imputation

K Nearest Neighbor for imputation

Responden	Q1	Q2	Q3	Q4	Q5
R1	2	3	4	2	1
R2	2		4	2	5
R3			2	4	
R4	2				

Distance between (R2, *)		Distance between (R3, *)		Distance between (R4, *)	
E(R2,R1)	4	E(R3,R1)	2.828427	E(R4,R1)	0
E(R2,R3)		E(R3,R2)	2.828427	E(R4,R2)	0
E(R2,R4)		E(R3,R4)		E(R2,R4)	

- Suppose we want to fill R3-Q1
 - **Complete case strategy:** A case can act as a neighbor if and only if it contains values for all attributes. Only R1 is a neighbor to R3.
 - **Incomplete case strategy:** A case can act as a neighbor if and only if it contains values for all attributes that the case being imputed has values for, and for the attribute being imputed. Both R1 and R2 are neighbor.
 - Once the strategy is picked. Sort the distances in ascending order. Pick the top K observations. Take the mean or median of the values for the variable being imputed. Fill the missing value.
 - Value of k:
 - Square root of number of observations N.
 - $k = 2$; $k = 1$ is sensitive to outliers.
 - $k = 10$ for large data sets.
 - As k increases, the mean distance to the donors gets larger, which implies that the replacement values could be less precise.
 - As k approaches N, the method converges to ordinary mean imputation.

Cluster Quality

Quality of Cluster

The Euclidian distance between any two observations within the cluster will be lesser than the observations between clusters. This is used to derive ideal number of clusters and quality of clusters

Elbow Method

- For each value of cluster C, calculate **within cluster sum of squared** (WCSS) distance.
- WCSS is the sum of squared distance between each point of the cluster and the centroid of the cluster.
- WCSS is maximum for K =1. For 2 Cluster with centroid C1 and C2, WCSS is:

$$\sum_{i=1, p_i \text{ in } c_1}^n d(p_i, C1) + \sum_{i=1, p_i \text{ in } c_2}^m d(p_i, C2)$$

- d(.) is the Euclidian distance of points from cluster centroid.

If the delta difference in WCSS between Cluster K to Cluster K+1 is small, then the optimal cluster size is K.

Silhouette Score

- Let a(i) be the average distance between an observation i and other points in the cluster 1 to which observation i belongs.
- Let b(i) be the minimum average distance between observation i and observations in other clusters (say cluster 2 and 3). We take the min of the averages.

$$S(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

- This is calculated for all the observations in a cluster and then the average silhouette score is reported for the cluster. Its value ranges from -1 to 1.
 - Score = 1: Means clusters are well apart from each other and clearly distinguished.
 - Score = 0: Means clusters are indifferent, or we can say that the distance between clusters is not significant.
 - Score = -1: Means clusters are assigned in the wrong way.

Silhouette Score - Interpretation

- Silhouette coefficients (as these values are referred to as) near +1 indicate that the sample is far away from the neighboring clusters.
- A value of 0 indicates that the sample is on or very close to the decision boundary between two neighboring clusters.
- A negative values indicate that those samples might have been assigned to the wrong cluster.

Distance measures

Manhattan distance

Assume that the data has two beer brands X_1 and X_2 with n -dimension features $X_1 (x_{11}, x_{12}, \dots, x_{1n})$ and $X_2 (x_{21}, x_{22}, \dots, x_{2n})$. Then the distance between X_1 and X_2 can be calculated as

$$DM(X_1, X_2) = \sum_{i=1}^n |X_{1i} - X_{2i}|$$

- Useful while measuring distance between different locations (for example distance between two shops in a city).

Manhattan Distance

Manhattan distance from Budweiser to all other beer brands

ID	BEER	CAL	SOD	ALC	COST
1	Budweiser	144	15	4.7	0.43
2	Schlitz	151	19	4.9	0.43
3	Lowenbrau	157	15	4.9	0.48
4	Kronenbourg	170	7	5.2	0.73
5	Heineken	152	11	5	0.77
6	Old Mil	145	23	4.6	0.28
7	Augsburger	175	24	5.5	0.4
8	Strohs	149	27	4.7	0.42
9	Miller lite	99	10	4.3	0.43
10	Bud light	113	8	3.7	0.44
11	Coors	140	18	4.6	0.44
12	Coors lite	102	15	4.1	0.45
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16	Pabst	68	15	2.3	0.38
17	Hamms	136	19	4.4	0.43
18	Heilemans	144	24	4.9	0.43
19	Olympia	72	6	2.9	0.46
20	Schilitz lite	97	7	4.2	0.47

Original Data

ID	BEER	CAL	SOD	ALC	COST	Total
1	Budweiser	0	0	0	0.000	0.000
2	Schlitz	7	4	0.2	0.000	11.200
3	Lowenbrau	13	0	0.2	0.050	13.250
4	Kronenbourg	26	8	0.5	0.300	34.800
5	Heineken	8	4	0.3	0.340	12.640
6	Old Mil	1	8	0.1	0.150	9.250
7	Augsburger	31	9	0.8	0.030	40.830
8	Strohs	5	12	0	0.010	17.010
9	Miller lite	45	5	0.4	0.000	50.400
10	Bud light	31	7	1	0.010	39.010
11	Coors	4	3	0.1	0.010	7.110
12	Coors lite	42	0	0.6	0.020	42.620
13	Michelob light	9	4	0.5	0.070	13.570
14	Becks	6	4	0	0.330	10.330
15	Kirin	5	9	0.3	0.360	14.660
16	Pabst	76	0	2.4	0.050	78.450
17	Hamms	8	4	0.3	0.000	12.300
18	Heilemans	0	9	0.2	0.000	9.200
19	Olympia	72	9	1.8	0.030	82.830
20	Schilitz lite	47	8	0.5	0.040	55.540

Manhattan distance from Budweiser

These distances are to be calculated for each beer brand



The total column is called the Manhattan distance

$$D(X_1, X_2) = \text{abs}(x_{11} - x_{21}) + \text{abs}(x_{12} - x_{22}) + \dots + \text{abs}(x_{1n} - x_{2n})$$

Minkowski distance

$$\text{Minkowski } D(X_1, X_2) = \left(\sum_{i=1}^n |X_{1i} - X_{2i}|^p \right)^{1/p}$$

- It is a generalized distance measure between two cases in the dataset.
- When $p = 1$, Minkowski distance is same as the Manhattan distance and
- When $p = 2$, Minkowski distance is same as the Euclidian distance.

Minsowski Distance

Minsowski distance from Budweiser to all other beer brands

ID	BEER	CAL	SOD	ALC	COST
1	Budweiser	144	15	4.7	0.43
2	Schlitz	151	19	4.9	0.43
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Original Data

ID	BEER	CAL	SOD	ALC	COST	Total
1	Budweiser	0	0	0	0.000	0.000
2	Schlitz	343	64	0.008	0.000	7.411
3	Lowenbrau	2197	0	0.008	0.000	13.000
4	Kronenbourg	17576	512	0.125	0.027	26.250
5	Heineken	512	64	0.027	0.039	8.321
6	Old Mil	1	512	0.001	0.003	8.005
7	Augsburger	29791	729	0.512	0.000	31.251
8	Strohs	125	1728	0	0.000	12.283
9	Miller lite	91125	125	0.064	0.000	45.021
10	Bud light	29791	343	1	0.000	31.119
11	Coors	64	27	0.001	0.000	4.498
12	Coors lite	74088	0	0.216	0.000	42.000
13	Michelob light	729	64	0.125	0.000	9.257
14	Becks	216	64	0	0.036	6.542
15	Kirin	125	729	0.027	0.047	9.488
16	Pabst	438976	0	13.824	0.000	76.001
17	Hamms	512	64	0.027	0.000	8.320
18	Heilemans	0	729	0.008	0.000	9.000
19	Olympia	373248	729	5.832	0.000	72.047
20	Schilitz lite	103823	512	0.125	0.000	47.077

Minsowski distance from Budweiser (p=3)

These distances are to be calculated for each beer brand



The total column is called the Minsowski distance

$$D(X_1, X_2) = \sqrt[p]{|x_{11} - x_{21}|^p + |x_{12} - x_{22}|^p + \dots + |x_{1n} - x_{2n}|^p}$$

Jaccard Similarity Coefficient (Jaccard Index)

$$J(X_1, X_2) = \frac{n(X_1 \cap X_2)}{n(X_1 \cup X_2)}$$

- Where $n(X_1 \cap X_2)$ is the number of observations in $(X_1 \cap X_2)$, $n(X_1 \cup X_2)$ is the number of observations in $(X_1 \cup X_2)$. Data has to be pre-processed to convert the data into binary or Boolean representation.
- If two customers have rated movies on similar scale (say both 5 star); then we can take the list of movies compare them and recommend the unseen movies to both. **Drawback??**
- **Jaccard distance = 1 – Jaccard similarity**

Text	nothing	is	impossible	the	word	itself	says	it	when	you	have	dream	you've	got	to	grab
nothing is impossible the word itself says it	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0
when you have dream you've got to grab it	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1

$$J(T1,T2) = 1/16 = 0.0625$$

Cosine Similarity

$$\text{Cosine_Similarity}(X_1, X_2) = \frac{X_1 \bullet X_2}{\|X_1\| \bullet \|X_2\|} = \frac{\sum_{i=1}^n X_{1i} \times X_{2i}}{\sqrt{\sum_{i=1}^n X_{1i}^2} \times \sqrt{\sum_{i=1}^n X_{2i}^2}}$$

- A lower cosine similarity indicates low similarity between two observations. Completely similar observations will have a cosine similarity value of 1.

Text	nothing	is	impossible	the	word	itself	says	it	when	you	have	dream	you've	got	to	grab
nothing is impossible the word itself says it	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0
when you have dream you've got to grab it	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1

$$C(T1,T2) = \frac{1*0+1*0+1*0+1*0+1*0+1*0+1*0+1*0+1*0+1*0+1*0+1*0+1*0+1*0+1*0+1*0+1*0}{\sqrt{1^2+1^2+1^2+1^2+1^2+1^2+1^2+1^2}*\sqrt{1^2+1^2+1^2+1^2+1^2+1^2+1^2+1^2}} = 1/(\sqrt{8} * \sqrt{9}) = 0.117$$



Represent document/sentences as vector of words. In vector space, if documents are similar the angle between the vectors will be less and cosine of angle will increase (Cos 0 = 1).

Gowers distance

- Used with dataset which has both qualitative and quantitative variable.

$$D_{ij} = \frac{\sum_{k=1}^n D_{ijk} * W_{ijk}}{\sum_{k=1}^n W_{ijk}}$$

D_{ijk} : Similarity between observation i and j for k th variable. Distance is $1 - D_{ijk}$. For numeric and ordinal variable:

$$D_{ijk} = 1 - \frac{|X_{ik} - X_{jk}|}{R_k}$$

X_{ik} : is value of observation i for variable k . R_k is the range of variable k .

For categorical variable:

$$D_{ijk} = \begin{cases} 1 & \text{when } X_{ik} = X_{jk} \\ 0 & \text{otherwise} \end{cases}$$

W_{ijk} : Binary variable. 1 when both observation i and j for k th variable is known. 0 if any one is missing.

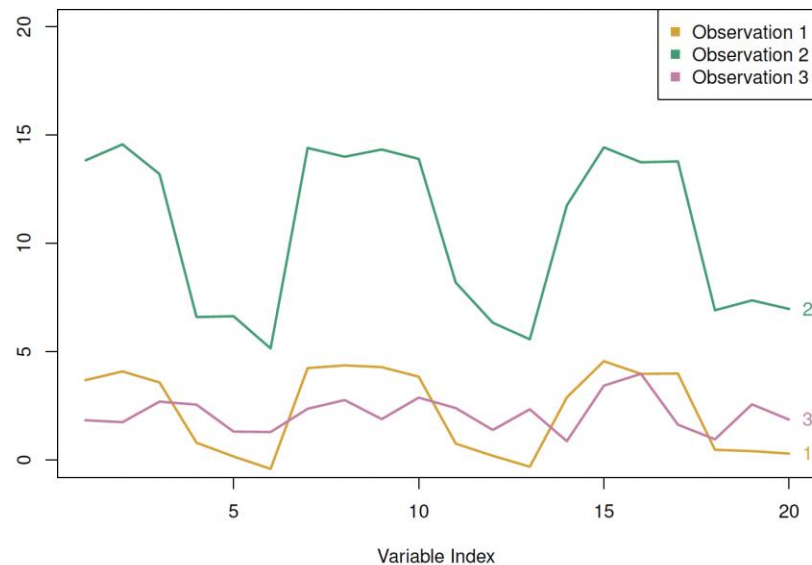
Gower Distance Calculation

Customer	Number of Movies downloaded					
	Action (k = 1,num)	Romance (k = 2, num)	Comedy (k = 3,num)	Sci-fi (k = 4,num)	Maximum Rating (k = 5, ordinal)	Married (k = 6, cat)
1	23	5	15	0	4	0
2	5	18	16	2	5	1
3	25	0	0	15	5	0
4	2	30	15	0	4	1
5	45	0	0	10	5	0
Range	43	30	16	15	1	

	Similarity of Customer 1 from customer 2						G Distance	G Similarity
Dijk(C12)	0.581	0.567	0.938	0.867	0.000	0	0.508	0.492
Wijk(C12)	1	1	1	1	1	1		
The first calculation between Cust 1 and Cust 2 for Action is $1 - ((\text{abs}(23-5)/43))$								
Customer	Similarity of Customer 1 from all customers						G Distance	G Similarity
D11	1.000	1.000	1.000	1.000	1.000	1	0.000	1.000
W11	1	1	1	1	1	1		
D12	0.581	0.567	0.938	0.867	0.000	0	0.508	0.492
W12	1	1	1	1	1	1		
D13	0.953	0.833	0.063	0.000	0.000	1	0.525	0.475
W13	1	1	1	1	1	1		
D14	0.512	0.167	1.000	1.000	1.000	0	0.387	0.613
W14	1	1	1	1	1	1		
D15	0.488	0.833	0.063	0.333	0.000	1	0.547	0.453
W15	1	1	1	1	1	1		

Correlation based distance

- Two observations are similar when their features are highly correlated, even if the observed values are far off in Euclidian distance.
 - Computed based on observation profile for each pair of observation (rather than on variables).



- Clustering customers based on past shopping history:
 - Rows represent shopper and columns represent #items purchased.
 - What type of distance measure should be used? [Euclidian vs. correlation?]

Density based spatial clustering of application with noise

Types of clustering algorithm

Clustering algorithms can be of two types: Partitioning (ex: k-means), hierarchical (ex: hierarchical clustering)

Partitioning algorithms

- Construct a partition of a dataset D of n objects into a set of k clusters.
 - Starts with an initial partition of D and then uses an iterative control strategy to optimize an objective function.
 - Each cluster is represented by the gravity center of the cluster (k-means algorithms) or by one of the objects of the cluster located near its center (k-medoid algorithms).
- Two step procedure
 - First, determine k representatives, minimizing the objective function.
 - Second, assign each object to the cluster with its representative “closest” to the considered object.
- k is an input parameter for these algorithms, i.e., some domain knowledge is required.

Hierarchical algorithms

- Create a hierarchical decomposition of dataset. The hierarchical decomposition is represented by a dendrogram, a tree that iteratively splits dataset into smaller subsets until each subset consists of only one object.
- The dendrogram can either be created from the leaves up to the root (**agglomerative approach**) or from the root down to the leaves (**divisive approach**) by merging or dividing clusters at each step.
- Do not need k as an input but termination condition has to be set to define when merge or division should be stopped.
- Say, in agglomerative approach, by using min distance between all the clusters.
 - What should be the min distance that is small enough to separate all “natural” clusters and at the same time large enough such that no cluster is split into two parts.

DBSCAN

- Ester et.al (1996) originally developed to manage spatial data (data related to space) – satellite data.
 - In earth observation database to discover classes of houses along some river.
 - It is based on agglomerative clustering (Hierarchical, single linkage) approach.
- Existing clustering techniques:
 - Does not scale to large data
 - Cannot discover cluster of arbitrary shapes
 - Depends on user defined parameters which needs domain expert
- It was devised as an improvement over another clustering algorithm – Clustering Large Applications based on RANdomized Search (CLARANS)



DBSCAN requires only one input parameter and supports the user in determining an appropriate value for it

Intuition of Density based clustering

- Finding clusters in the world of data is similar to discovering islands in natural world.
 - Region of data with higher density (land mass) is surrounded by noise (sea).
 - There could be several land masses (higher density region) separated by sea (noise).
- The idea behind clustering is to separate the regions of high density from the sparsely populated data (sea). So cluster is:
 - **Highly dense region separated by sparse region** – Probability density function perspective.
 - **Highly probable region separated by improbable region** - This is the view from a random sample drawn from the distribution.



Definitions to understand DBSCAN

- Let us set $\varepsilon = 0.5$ and $MinPts = 5$ (think of them as parameters)
- Pick a point p at random from the dataset D
- **Def 1:** Calculate the epsilon neighborhood of point P called as $|N_{eps}(p)|$
 - We can use any distance measuring technique (say, Euclidian distance). **A point q is a neighbor of point p if,**

$$d(p, q) \leq \varepsilon$$

- But we only need neighbor of core points. So for p to be a core point, it should have at least $MinPts$ in the vicinity of point p

$$|N_{eps}(p)| \geq MinPts$$

- Think of p as the seed value (core of the cluster) from where further expansion of the cluster will happen.
- All the points within the neighbor of p are said to be directly density reachable (DDR)

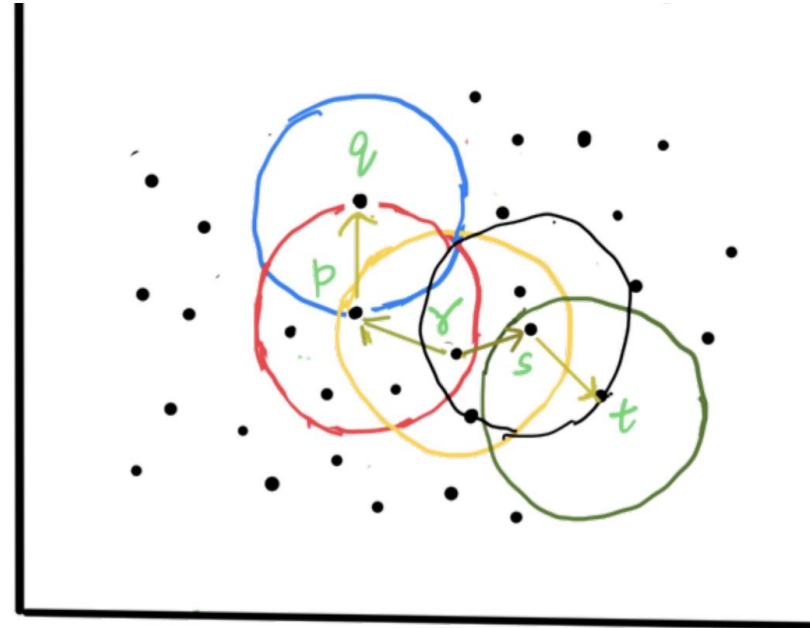
This definition of neighbor manifests into several metrics

Directly Density Reachable

- **Def 2:** A point q is directly density reachable from point p if

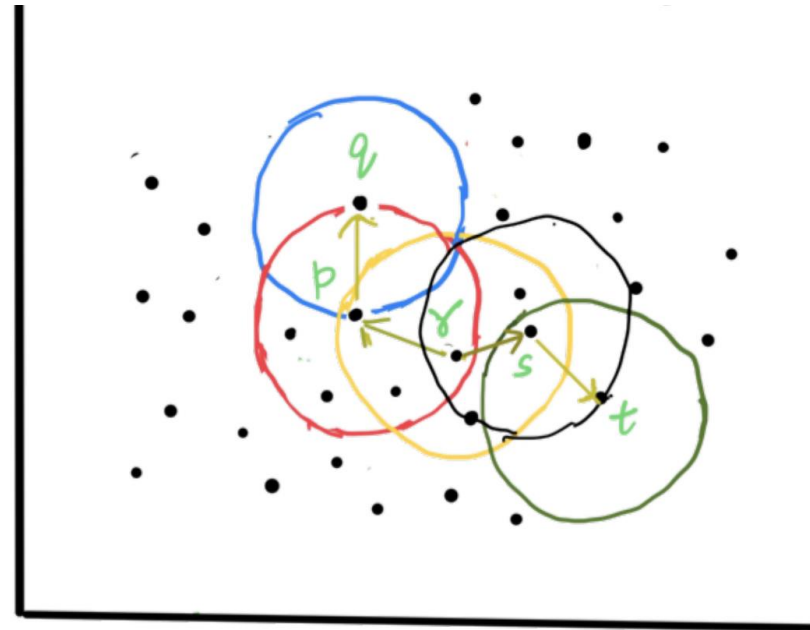
$$q \in N_{eps}(p) \\ |N_{eps}(p)| \geq MinPts \text{ (Core point condition)}$$

- DDR is symmetric for the core points but not for border points (say q).
 - q is DDR from p but p is not DDR from q .



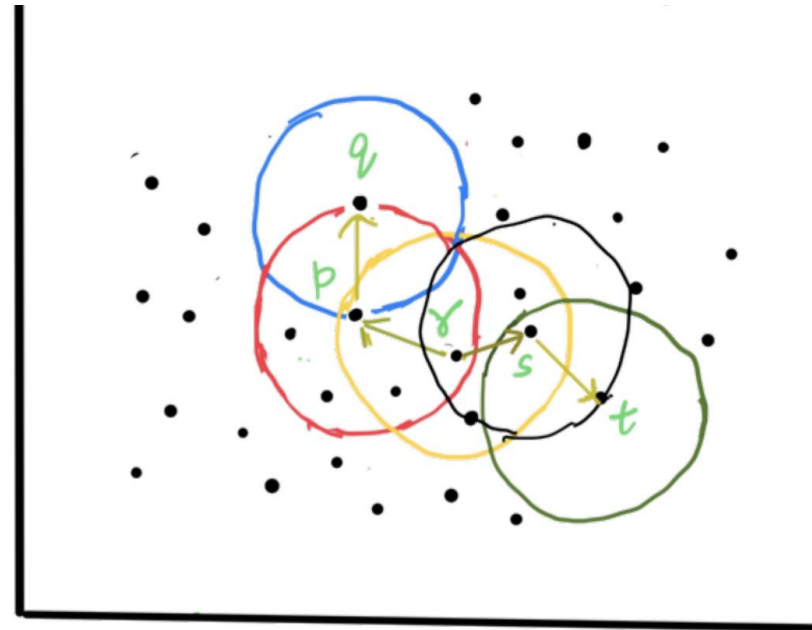
Density Reachable

- **Def 3:** A point q is density reachable from point r if there is chain of points such that the intermediate points are directly density reachable.
- DR is symmetric for the core points but not for border points (say q).
 - q is DDR from r as there is point p (which is DDR from r) to reach q .
 - But r is not DR from q .
- Two border points of the same cluster may not be DR as core point condition for them will not hold. So we need another definition.



Density Connected

- **Def 4:** A point q is density connected from point t as both of these points are density reachable from r .
- DC is symmetric as well as reflexive
 - q is DC to t as there is point r to which they are DR.
 - t is DC from q as there is point r to which they are DR.



What the definitions achieve?

Def 1 and Def 2 define set of points that are **core points**.

Def 3 and Def 4 **expands the boundary** from a given core point to include other points in the vicinity.

These 4 definitions help **build a cluster**.

DBSCAN – How it works

Two step procedure:

- Given the parameter epsilon and MinPts; identify points from the dataset which satisfy core point condition.
 - Intuitively, think of a list of such points which is created in this step
 - Not every point will be a core point
- Retrieve all the points that are density reachable or density connected to the first seed point.
 - If p belongs to cluster C and q is density reachable or density connected then (p,q) belongs to same cluster.

The value of epsilon and MinPts from the thinnest cluster are the best candidate for setting the global value of epsilon and MinPts. Heuristics can be used to find them.

Limitations??



https://hdbscan.readthedocs.io/en/latest/comparing_clustering_algorithms.html#dbscan
https://scikit-learn.org/stable/auto_examples/cluster/plot_dbscan.html

DBSCAN Limitations

- DBSCAN can only provide a flat (non hierarchical) clustering
 - as it uses a global epsilon and MinPts. These are critical input parameters:
 - Clusters of different threshold may get merged or split due to global threshold.
- DBSCAN and many other clustering technique uses single linkage strategy which is highly influenced by noise.
 - A single data point at wrong place will act as a bridge to connect two clusters.

HDBSCAN Clustering

HDBSCAN

HDBSCAN (Campello, 2013) is a complete density-based clustering hierarchy from which a simplified hierarchy composed only of the most significant clusters can be easily extracted:

- Does not use global threshold values of parameters.
- Robust to noise (sea level) as uses novel approach called mutually reachable distance to spread the noise.
- Improvement over DBSCAN and OPTICS algorithm

Key definitions or concepts governing HDBSCAN:

- **Core distance** calculated using KNN algorithm. This is called as core object (similar to core point in DBSCAN)
- **ϵ – core object is** defined based on threshold level ϵ . Epsilon is used to identify points which will be building block of cluster.
- **Mutually/symmetric reachable distance (MRD)** based on Euclidian distance for computing distance. It is used to connect the ϵ – *core object*.
- **Transformed proximity graph OR mutually reachable graph (MRG)** – similar to distance matrix but filled with MRD.

Core Distance

- Set $m_{pts} = 6$ to get 6 nearest neighbor to point p (including p) for every point P in the dataset D .
 - m_{pts} is a user defined input similar to MinPts in DBSCAN.

Def 1: Define core distance

- Def 1: For every point p in dataset D , draw a circle of radius r_p such that there are K points within the circle. **This r_p is the core distance $d_{c_k}(p)$:**

$$d_{c_k}(p) = r_p \text{ (distance to the } 5 \text{ nearest neighbor)}$$

Core distances are an estimate of the density.



- The smaller the core distance of a point, the more dense region it belongs to (like core point in DBSCAN).
- The larger the core distance of a point, the more sparse region it belongs to (like border point in DBSCAN). Def 2, spreads these points to make single linkage robust to noise.

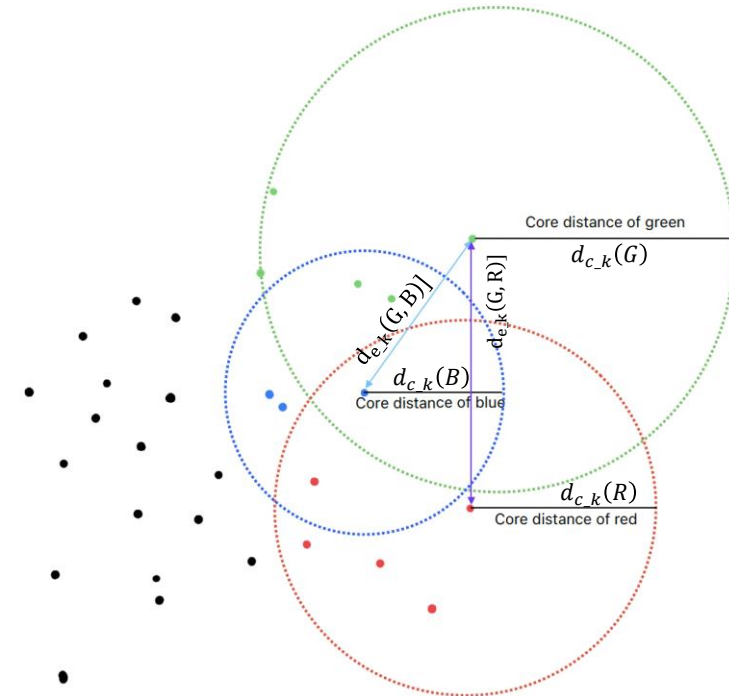
Calculate Mutual Reachability Distance

Def 2: Mutual Reachability Distance

- The mutual reachability distance (MRD) between blue and green circle will be equal to the core distance of green (radius of green circle).
 - The line joining center of blue and green circle passed through only the blue circle and not the green circle. This implies that core distance of green is maximum.

$$d_{m_k}(G, B) = \max[d_{c_k}(G), d_{c_k}(B), d_{e_k}(G, B)]$$

$$d_{m_k}(G, B) = d_{c_k}(G) \Rightarrow a \text{ dense region}$$



What is the mutual reachability between Green and Red?

Mutual Reachability Distance

Used to spread the points which are in sparse region:

- For two points p, q in the dataset D : we know $d_{c_k}(p) = r_p$ and $d_{c_k}(q) = r_q$. Now Say the Euclidian distance between p and q i.e. $d_{e_k}(p, q)$.
- So mutual reachability distance (MRD) is

$$d_{m_k}(p, q) = \max[d_{c_k}(p), d_{c_k}(q), d_{e_k}(p, q)]$$

MRD for two points (p, q) in dense region is same as one of the core distances.

What is the impact of MRD on points in sparse region?

Step 1: Mutual Reachable Graph

Def 3: Mutual Reachability Graph

- MRG ($G_{m_{pts}}$) is just like a distance matrix created in hierarchical clustering but with MRD as weights
 - For beer example, MRG will be a 20*20 matrix with MRD as edges weight.
 - The MRD will be either:
 - Core distance (distance to K nearest beer brands) or
 - Euclidian distance between the beer brands.
 - The diagonal is not set to 'zero' but as the core distance. This is used in extension of MST to extract the hierarchy from HDBSCAN.

	1	2	3	...	20
1	0	MRD	MRD	...	MRD
2	MRD	0	MRD	...	MRD
3	MRD	MRD	0	...	MRD
•				...	MRD
•				...	MRD
•				...	MRD
20	MRD	MRD	MRD	...	0

ϵ – core object

Def 4: ϵ – core object

- Pick a value of epsilon (ϵ) and compare it with the core distances of each point. If core distance $> \epsilon$, then remove those weights.
- The remaining vertices are ϵ –core object which intuitively gives the building block for disconnected cluster.

Point p will be ϵ – core object for every value of ϵ greater than or equal to core distance of p :

$$\text{for } \epsilon \geq d_{c_k}(p); p \text{ is } \epsilon - \text{core}$$

Think of epsilon as sea level which we can increase or decrease.

Cluster formation and ε – core object

Clusters are connected component of ε – core object.

We have identified core objects. But we need hierarchy as well i.e. how the nodes get connected.



Range of Epsilon $[0, \infty)$:

- Value close to 0 will imply everything is connected.
- If we start with a high value and steadily lower, we drop different edges. This results in disconnected nodes.

Intuition of how HDBSCAN works

We have single linkage strategy as we know from K-means. So steps are:

1. Connect the two components/objects with the minimum distance (identified by KNN or MRD) in MRG matrix.
 2. We have a cluster where two points are connected. Recalculate the MRG and go to step 1.
 3. Stop when all the objects in the cluster are connected.
- The single linkage hierarchy (dendrogram) on MRG can be processed further to identify connected components and noise objects at each level:
 - If core distance $>$ epsilon, remove them from the dendrogram and flag them as noise.
 - Single linkage on MRG decides how the hierarchy will be formed.
 - Core distance with epsilon decides which nodes will be core objects (retained in dendrogram).

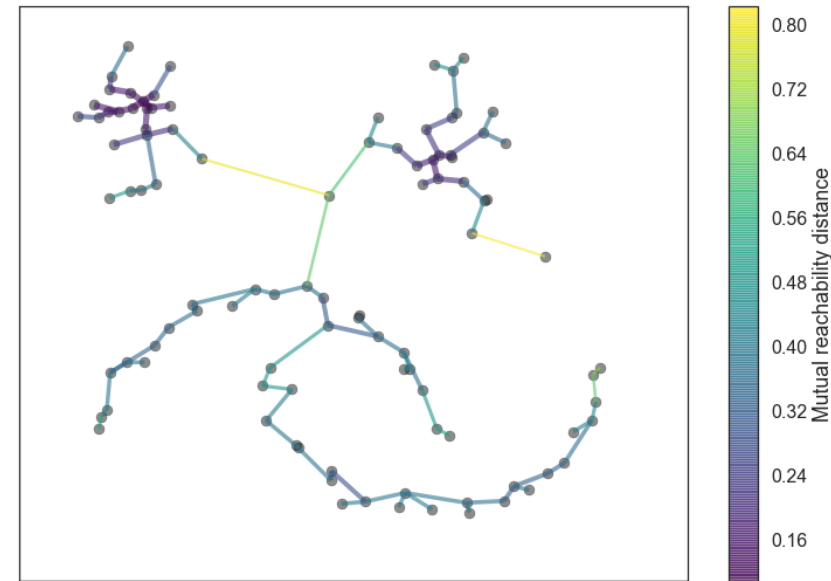
But the above will be a costly operation. So Minimal Spanning Tree is used which is equivalent to single linkage.

Step 2: Minimal Spanning tree

MST connects all the points in such a way that the sum of the edge weights (in this case, mutual reachability distances) is minimized.

- Find a minimal set of edges such that dropping any edge from the set causes a disconnection of components.
- We need the edge such that there is lowest weight edge connecting the components.

Prims or Kruskal's algorithm can be used.



- https://hdbscan.readthedocs.io/en/latest/how_hdbscan_works.html – for more detail.
- <https://pberba.github.io/stats/2020/01/17/hdbscan/> – to build intuition.

Step 3: Build hierarchy

Leaf node of the MST merge, nodes merge based on smallest edge weight—i.e., mutual reachability distance—in the MST.

- sort the edges of the tree in increasing order of distance (MRD) and then iterate through, creating a new merged cluster for each edge.
- merge happen based on single linkage strategy. **This creates a dendrogram.**

Step 4: Condense cluster

Condense Cluster:

Some additional parameters to simplify:

- A connected component is deemed spurious if it has fewer than m_{clSize} (minimum cluster size) objects.
 - At each split of the dendrogram, if cluster has $<$ minimum cluster size; do not split the tree else split.
- $m_{clSize} = 1$ is an isolated node
- Generally m_{clSize} is set equal to m_{pts} .
- The data points are labeled based on the cluster it belongs.

Difference:

DBSCAN creates flat cluster by drawing a single horizontal line through dendrogram and any cluster with single data point is considered noise.

Applying Unsupervised Learning Algorithm to detect fraud

Context

1 >

Earnings management occurs when managers use their judgment in financial reporting and in structuring transactions to alter financial reports to mislead stakeholders about the underlying economic performance of the company

2 >

As on December 2016, the total number of listed companies in Indian stock exchange was approximately 5622.

3 >

Securities and Exchange Board of India (SEBI) reported that approximately 3.14% of the Indian companies are involved in earnings manipulation.

Financial Ratios

Days Sales to Receivables Index (DSRI)

$$DSRI = \frac{\frac{Receivable_{(t)}}{Sales_{(t)}}}{\frac{Receivable_{(t-1)}}{Sales_{(t-1)}}}$$

DSRI greater than 1 implies revenue inflation

Gross Margin Index (GMI)

$$GMI = \frac{\frac{Sales_{(t-1)} - Cost\ of\ Goods\ Sold_{(t-1)}}{Sales_{(t-1)}}}{\frac{Sales_{(t)} - Cost\ of\ Goods\ Sold_{(t)}}{Sales_{(t)}}}$$

GMI greater than 1 means gross margin is deteriorating

Asset Quality Index (AQI)

$$AQI = \frac{\frac{1 - (Current\ Asset_{(t)} + netPPE_{(t)})}{Total\ Assets_{(t)}}}{\frac{1 - (Current\ Asset_{(t-1)} + netPPE_{(t-1)})}{Total\ Assets_{(t-1)}}}$$

AQI greater than 1 may indicate the tendencies of capitalizing and deferring costs that should have been expensed

Sales Growth Index (SGI)

$$SGI = \frac{Sales_{(t)}}{Sales_{(t-1)}}$$

SGI greater than or less than 1 may indicate that the firm is under possible pressure to manipulate earnings to keep up appearances

Financial Ratios

Depreciation Index (DEPI)

$$DEPI = \frac{\frac{Depreciation Expense_{(t-1)}}{(Depreciation Expense_{(t-1)} + netPPE_{(t-1)})}}{\frac{Depreciation Expense_{(t)}}{(Depreciation Expense_{(t)} + netPPE_{(t)})}}$$

DEPI greater than 1 may indicate tendencies of the assets being depreciated at a slower rate to boost earnings

Sales and General Administrative (SGAI)

$$SGAI = \frac{\frac{SGAExpense_{(t)}}{Sales_{(t)}}}{\frac{SGAExpense_{(t-1)}}{Sales_{(t-1)}}}$$

SGAI less than 1 may indicate that the company may manipulate earnings to defer costs

Accruals to Asset Ratio (ACCR)

$$ACCR = \frac{Profit after Tax_{(t)} - Cash from Operations_{(t)}}{Total Assests_{(t)}}$$

ACCR greater than 1 may indicate that the accruals can possibly be used to manipulate earnings

Leverage Index (LEVI)

$$LEVI = \frac{\frac{(LTD_{(t)} + CurrentLiabilities_{(t)})}{Total Assests_{(t)}}}{\frac{(LTD_{(t-1)} + CurrentLiabilities_{(t-1)})}{Total Assests_{(t-1)}}}$$

Data Collection

Data Collection:

- SEBI and Prowess database used as the source to collect data about public listed company in India.
- Out of 1239 observations, 39 observations which were identified as doing fraud by SEBI in earlier audits

Train and test set:

- 70% of the data used for model training (840 Non manipulators + 28 Manipulators = 868 observations). 5 cross validation set created for the 868 observations
- 30% of the data used for model testing (360 Non manipulators + 11 Manipulators = 371 observations)

Multivariate Gaussian Model

- Identify the features
 - 1239 observations across eight financial ratios
 - 1200 ($Y = 0$; *non manipulators*) and 39 ($Y = 1$; *manipulators*)
 - *training set* = $\{x^{(1)}, x^{(2)}, \dots, x^{(m)}\}$ where $m = 900$ obs. across eight ratios
 - *test set* = $\{(x^1_{test}, y^1_{test}), (x^2_{test}, y^2_{test}), \dots, (x^m_{test}, y^m_{test})\}$ where $m = 339$ obs. across eight ratios (300 non manipulators and 39 manipulators)
- Compute mean and standard deviation for all the features in train set
- Given a test set compute

$$P(x) = \prod_{j=1}^n (1/\sqrt{2\pi} \sigma_j) * \exp\left(-\frac{(x_j - \mu_j)^2}{2\sigma_j^2}\right) \text{ where } n \text{ is the number of feature}$$

$$Y = \begin{cases} 1 & \text{if } P(x) < \epsilon \text{ then anomaly} \\ 0 & \text{If } P(x) \geq \epsilon \text{ then ok} \end{cases}$$

- Fine tune ϵ to catch anomaly

Outcome of Multivariate Gaussian models

Classification Matrix at ϵ 0.25

		Count of Predicted Status		
Actual Count		No	Yes	Total
	No	214	86	300
	Yes	10	29	39
	Grand Total	224	115	339

Model Performance on Company Financial Ratios

Model (70% Train, 30% Test)	Performance on Test Set (11- Yes, 339 – No)			ROC on Test Set
	Sen	Spec	Overall	
Gaussian Model	0.71	0.74	0.72	

Sensitivity = Non-Manipulator (No), Specificity = Manipulator(Yes)

Principal component analysis

USArrest Dataset

USArrest dataset: Number of arrests per 100, 000 residents for each of three crimes and % of Urban Pop										Principal Components			
Sates	Murder	Assault	UrbanPop	Rape	States	Murder	Assault	UrbanPop	Rape	Z1	Z2	Z3	Z4
Alabama	13.2	236	58	21.2	Montana	6	109	53	16.4				
Alaska	10	263	48	44.5	Nebraska	4.3	102	62	16.5				
Arizona	8.1	294	80	31	Nevada	12.2	252	81	46				
Arkansas	8.8	190	50	19.5	New Hamp	2.1	57	56	9.5				
California	9	276	91	40.6	New Jersey	7.4	159	89	18.8				
Colorado	7.9	204	78	38.7	New Mexico	11.4	285	70	32.1				
Connecticu	3.3	110	77	11.1	New York	11.1	254	86	26.1				
Delaware	5.9	238	72	15.8	North Caro	13	337	45	16.1				
Florida	15.4	335	80	31.9	North Dakc	0.8	45	44	7.3				
Georgia	17.4	211	60	25.8	Ohio	7.3	120	75	21.4				
Hawaii	5.3	46	83	20.2	Oklahoma	6.6	151	68	20				
Idaho	2.6	120	54	14.2	Oregon	4.9	159	67	29.3	For representation purpose			
Illinois	10.4	249	83	24	Pennsylvan	6.3	106	72	14.9				
Indiana	7.2	113	65	21	Rhode Islar	3.4	174	87	8.3				
Iowa	2.2	56	57	11.3	South Caro	14.4	279	48	22.5				
Kansas	6	115	66	18	South Dakc	3.8	86	45	12.8				
Kentucky	9.7	109	52	16.3	Tennessee	13.2	188	59	26.9				
Louisiana	15.4	249	66	22.2	Texas	12.7	201	80	25.5				
Maine	2.1	83	51	7.8	Utah	3.2	120	80	22.9				
Maryland	11.3	300	67	27.8	Vermont	2.2	48	32	11.2				
Massachus	4.4	149	85	16.3	Virginia	8.5	156	63	20.7				
Michigan	12.1	255	74	35.1	Washington	4	145	73	26.2				
Minnesota	2.7	72	66	14.9	West Virgir	5.7	81	39	9.3				
Mississippi	16.1	259	44	17.1	Wisconsin	2.6	53	66	10.8				
Missouri	9	178	70	28.2	Wyoming	6.8	161	60	15.6				

How PCA helps

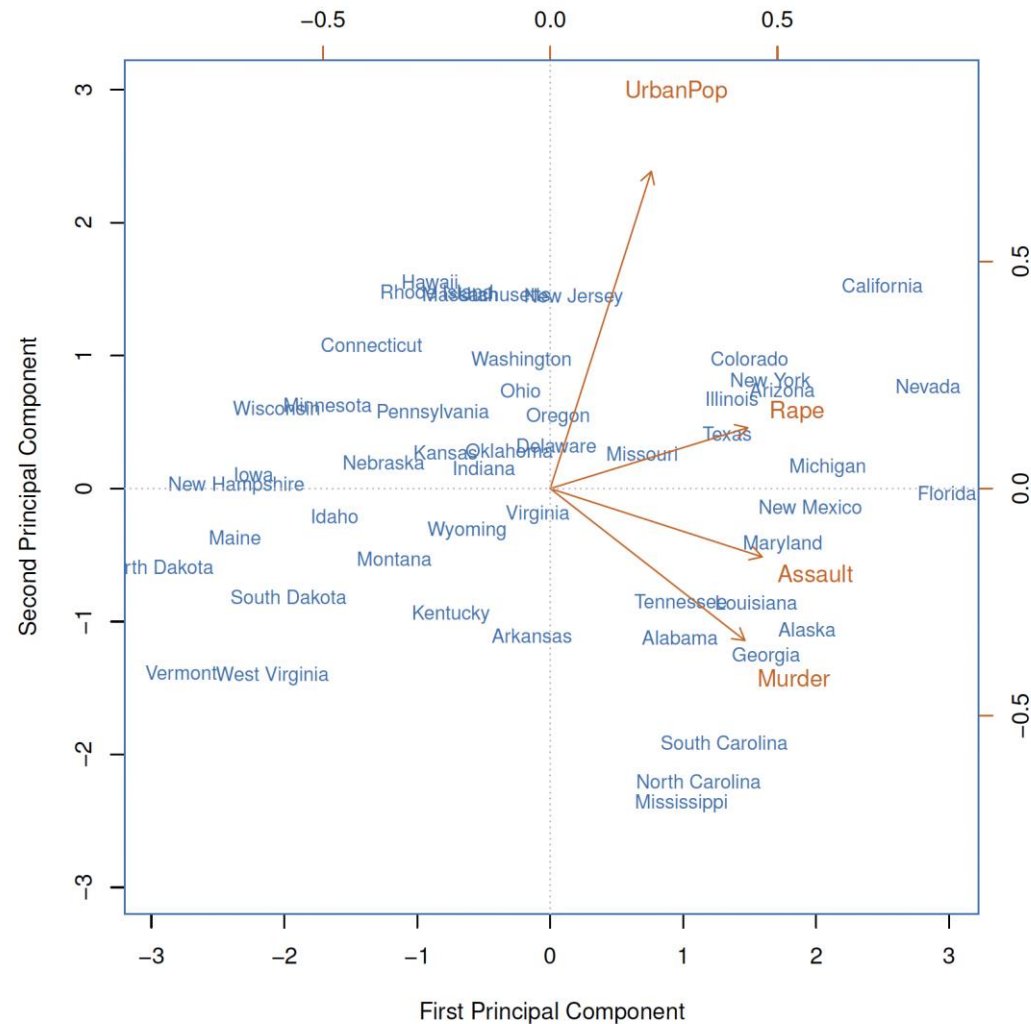
- Standardizing each variable to have mean zero and standard deviation of one.

Loading vectors (Secondary axis)

- PC1 has high and equal weight for crimes but less weight for urban population. This component measures serious crime.
 - crime-related variables are correlated. States with high murder rates tend to have high assault and rape rates
- PC2 has high weight on Urban Pop and thus measures the level of urbanization.

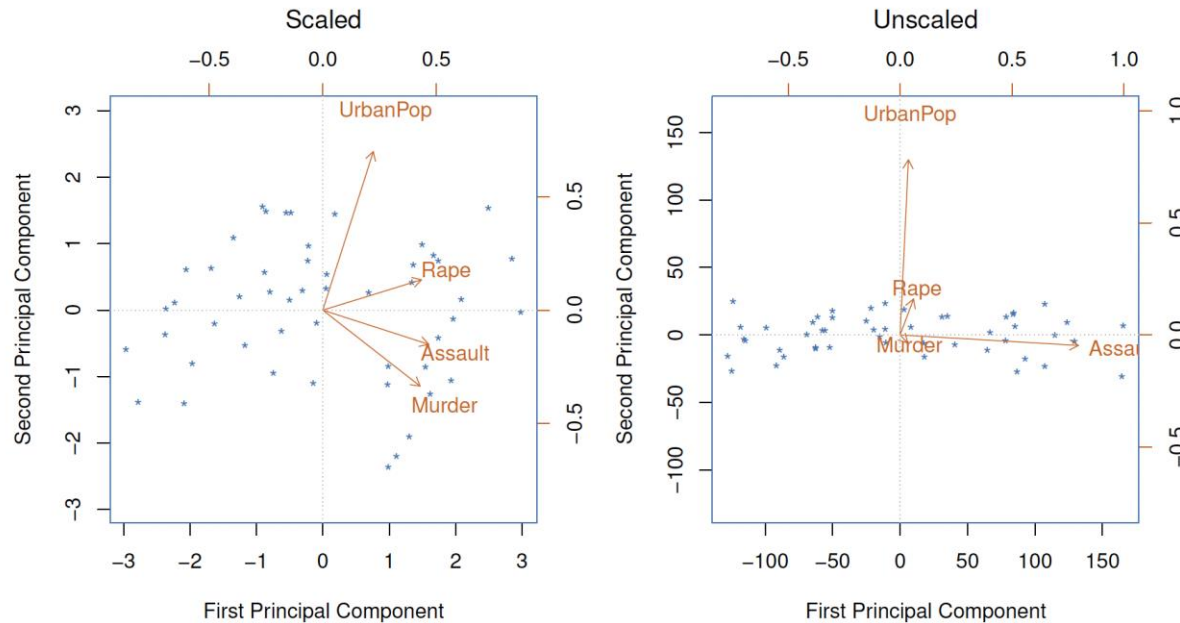
Loading score (Primary axis)

- California, Nevada and Florida, have high crime rates, while states like North Dakota, with negative scores on the PC1 has low crime rate.
- California has high score on PC2 as well. Implying high level of urbanization.



The blue state names represent the scores (n=50) for the first two principal components. The orange arrows indicate the first two principal component loading vectors (p = 4; with axes on the top and right).

Why scaling is needed?



- Different units of measurement and low/high variance of individual variables will have a major impact on the loading vector.
 - Murder, Rape, Assault and Urban Pop has variances: 18.97, 87.73, 6945.16, and 209.5
 - If not scaled; PC1 loading vector will have a very large loading for Assault.

Principal component analysis

- PCA refers to the **process by which the principal components** are computed and are subsequently used in understanding the data:
 - It can be used for data visualization in a lower dimensional space.
 - It can be used in supervised learning tasks.
 - It can be used for data imputation (through matrix completion) as well as for building recommender system.

What do we achieve with principal components?

Think of PCs as derived features.

- When there is a large set of correlated variables, **principal components** allow to **summarize** the set with a smaller number of representative variables that collectively **explain** most of the **variability** in the original set.
 - The principle components are directions in feature space along which the original data has high variability.

What are principal components

- Visualize n observations with measurements on a set of p features.
 - These are n observations living in a p -dimensional space.
 - Scatter plot? – If 10 features, we have 45 scatter plots.
 - Not informative as each plot contains small fraction of information present in the data.
- PCA finds a **lower dimensional representation** of the data that contains most of the variation from the original data.
- The **objective of PCA is to seek a small number of dimensions (PCs)** that are as interesting.
 - Interesting?? – amount by which the observations vary along each dimension.
 - The dimension along which the data varies the most, is most interesting.
 - Each of the dimensions found by PCA is a linear combination of the p features.

How is it computed – vector form

- Given an $n \times p$ data matrix, first each of the variables are centered to have a mean of zero and standard deviation of 1
 - Why?? Since we are only interested in maximizing variance along dimensions.
 - The loadings corresponding to variable with high variance will be higher.
- The **first PC** of a set of features $X_1, X_2 \dots X_p$ is the **normalized linear combination** of the features:

$$Z_1 = \varphi_{11} * X_1 + \varphi_{21} * X_2 + \dots + \varphi_{p1} * X_p$$

$$\text{s.t : } \sum_{j=1}^p \varphi_{j1}^2 = 1 \text{ (normalized means this constraint)}$$

Loading vectors

$\varphi_{11}, \varphi_{21}, \dots, \varphi_{p1}$ are the *loadings of the first principal component*. Together, $\varphi_1 = (\varphi_{11}, \varphi_{21}, \dots, \varphi_{p1})^T$ is called the *loading vector*.

- Intuitively, gives a sense of direction in feature space.
- φ_1 is a direction in feature space along which the data varies the most.
- If we project the n data points $x_1, x_2 \dots x_n$ onto this direction, the projected values are the principal component scores $z_{11}, z_{21} \dots z_{n1}$ themselves

First PC – in non vector form

The first PC of a set of features $X_1, X_2 \dots X_p$ is the normalized linear combination of the features:

$$z_{i1} = \varphi_{11} * x_{i1} + \varphi_{21} * x_{i2} + \dots + \varphi_{p1} * x_{ip}$$

$$\text{s.t : } \sum_{j=1}^p \varphi_{j1}^2 = 1 \text{ (normalized means this constraint)}$$

- $z_{11}, z_{21}, \dots z_{n1}$ are called the scores of first principal component. $Z_1 = (z_{11}, z_{21}, \dots z_{n1})^T$ is the score vector.

The objective: maximize Z_1 or the sample variance of n values of z_{i1} ($i = 1, 2 \dots n$):

$$\max_{\varphi_{11}, \varphi_{21} \dots \varphi_{p1}} \left\{ \frac{1}{n} \sum_{i=1}^n (z_{i1})^2 \right\} = \max_{\varphi_{11}, \varphi_{21} \dots \varphi_{p1}} \left\{ \frac{1}{n} \sum_{i=1}^n \left(\sum_{j=1}^p \varphi_{j1} * x_{ij} \right)^2 \right\}$$
$$\text{s.t: } \sum_{j=1}^p \varphi_{j1}^2 = 1$$

Second PC – in non vector form

The second PC of $X_1, X_2 \dots X_p$ is the **normalized linear combination** of the features that has maximal variance out of all linear combinations that are **uncorrelated with z_{i1}** :

$$z_{i2} = \varphi_{12} * x_{i1} + \varphi_{22} * x_{i2} + \dots + \varphi_{p2} * x_{ip}$$

$$\text{s.t: } \sum_{j=1}^p \varphi_{j2}^2 = 1 \text{ (Normalization constraint)}$$

$$\varphi_1 \cdot \varphi_2 = 0 \text{ (Orthogonality constraint)}$$

So, the objective: maximize the sample variance of n values of z_{i2} ($i = 1, 2 \dots n$):

$$\begin{aligned} \max_{\varphi_{12}, \varphi_{22} \dots \varphi_{p2}} \left\{ \frac{1}{n} \sum_{i=1}^n (z_{i2})^2 \right\} &= \max_{\varphi_{11}, \varphi_{21} \dots \varphi_{p1}} \left\{ \frac{1}{n} \sum_{i=1}^n \left(\sum_{j=1}^p \varphi_{j2} * x_{ij} \right)^2 \right\} \\ \text{s.t: } \sum_{j=1}^p \varphi_{j2}^2 &= 1 \\ \varphi_1 \cdot \varphi_2 &= 0 \end{aligned}$$

How do we compute the values of phi's

- Eigen decomposition or singular value decomposition is used to solve it.
 - The loading vectors i.e. the principal component directions $(\varphi_1, \varphi_2, \varphi_3, \dots)$ are the ordered sequence of eigenvectors of the matrix $X^T X$
 - The score vector i.e. variances of the components $(Z_1, Z_2 \dots Z_n)$ are the eigenvalues.

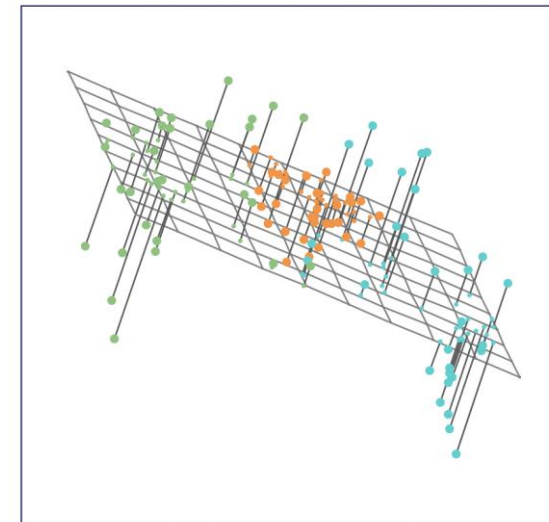
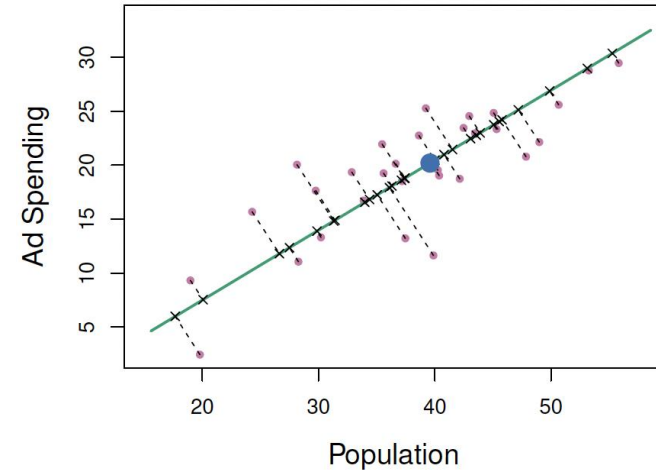
What do we achieve:

- We get the score vectors or the PCs: $Z_1, Z_2 \dots Z_n$. There are at most: $\min(n - 1, p)$ PCs.
 - Geometrically, this amounts to projecting the original data down onto the subspace spanned by $(\varphi_1, \varphi_2, \varphi_3, \dots)$, and plotting the projected points.

Intuition on PCs

- The first PC, **loading vector**, is a line in p -dimensional space that is closest to the n observations in the actual dataset.
- The first two principal components of a data set span the plane that is closest to the n observations.
- The first three principal components of a data set span the three-dimensional hyperplane that is closest to the n observations.

Closeness is measured in terms of average squared Euclidean distance.



Variance explained and Scree Plot

- No well defined objective on how many PCs are needed.
 - If using in supervised learning we can perform n fold CV to tune #PCs.

Proportion of Variance Explained (PVE)

1. Total variance in the data (when mean centered)

$$\sum_{j=1}^p \text{Var}(X_j) = \sum_{j=1}^p \frac{1}{n} \sum_{i=1}^n x_{ij}^2$$

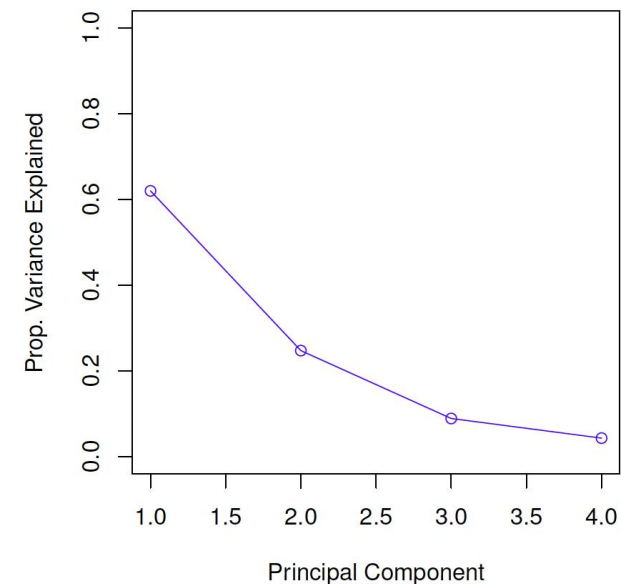
2. Variance explained by m^{th} PC

$$\frac{1}{n} \sum_{i=1}^n (z_{im})^2 = \frac{1}{n} \sum_{i=1}^n \left(\sum_{j=1}^p \varphi_{jm} * x_{ij} \right)^2$$

$$PVE = \frac{2}{1}$$

Scree Plot

- Pick the smallest number of PCs that are required in order to explain a sizable amount of variation in the data.



Application

Event Detection Data

- Year 2018 - 312, 142 records across 113 variables
 - Fire event - 40, Non Fire Event - 312, 102
- Year 2019 - 331, 057 records across 113 variables
 - Fire event - 20, Non Fire Event - 331, 037
- High dimensional data: Sensors measuring status, temp, airflow, chemical flow, chemical spray, pressure, speed, land scan, vacuum control, load at the belt etc.

Derived features

- Several business rules for creating derived features
 - Top and bottom Blade change event:
 - If top doctor status is unloaded and changes to loaded within next 5 mins, then blade change else not a blade change event.
 - 715 blade change event if the condition is exactly 5 minutes. 4 Fire events in these 715 events.
 - Felt Change event:
 - If (TM2_Cogen_WEHoodTemp_2120TC1.PV <= 800 and TM2_PIMS_Shift_AvgSpeed.L == 0)
 - Age of blade change and felt change
 - Top Blade Change Age = Time elapsed (in Hrs) since the last top blade change
 - Bottom Blade Change Age = Time elapsed (in Hrs) since the last bottom blade change
 - Felt Blade Change Age = Time elapsed (in Hrs) since the last felt blade change

- Other derived features:
 - $\text{Percentage_change_Kalipol_flow} = (\text{TM2_DE_ICON_CCR_Kalipol_Flow.PV} - \text{lag}(\text{TM2_DE_ICON_CCR_Kalipol_Flow.PV}, 5)) / \text{lag}(\text{TM2_DE_ICON_CCR_Kalipol_Flow.PV}, 5)$
 - $\text{percentage_change_in_metering_speed} = (\text{TM2_DE_ICON_CCR_Kalipol_Metering_Pump2_Speed.PV} - \text{lag}(\text{TM2_DE_ICON_CCR_Kalipol_Metering_Pump2_Speed.PV}, 5)) / \text{lag}(\text{TM2_DE_ICON_CCR_Kalipol_Metering_Pump2_Speed.PV}, 5)$
 - $\text{percentage_change_in_hood_temperature} = (\text{TM2_Cogen_WEHoodTemp_2120TC1.PV} - \text{lag}(\text{TM2_Cogen_WEHoodTemp_2120TC1.PV}, 5)) / \text{lag}(\text{TM2_Cogen_WEHoodTemp_2120TC1.PV}, 5)$
- Several derived ratios: 13 such features.
- Total of 134 features post feature engineering.

Several Challenges

- Only 60 positive cases/event of interest.
 - 0.0009% positive class
- High dimensional data (Sensors measuring status, temp, airflow, chemical flow, chemical spray, pressure, speed, land scan, vacuum control, load at the belt)
 - Data at minute level interval.
 - Mapping of fire events.
 - If(Fire Event ID = '', 'No Fire', 'Fire')
 - Mapping was inaccurate:
 - Top/bottom blade change event will follow the fire event (with lead of 1). However, this condition may not get satisfied if the fire event is not correctly marked in the data. At times, the lead time can be greater than 1 as well.
 - Instances where top/bottom blade change event was marked before the fire event but with a lag of 1,2,3 etc. This also suggest that the fire event mapping is incorrect.

Sanctity check on event mapping

- Select *Date*, *Top Doctor Status*, *Dryer Doctor Status*, *Top blade change event*, *Bottom blade change event*, *Fire Event Id*, *Event*:
 - where the record is marked as Fire event and the following record (lead of 1, 2, 3...) marked as a blade change event. Since it is first following record to Fire event, it will most likely be a No Fire event record.
 - Select the previous record (lag of 1,2,3...; Fire event) to this condition as well.

Table 4: Leading - Top Blade Change and Fire Event Mapping

Date_Time	TM2_DE_TopDoctorStatus_ZI_6860A.PV	Top_Blade_Change_Event	Fire_Event_Id	Event
2018-01-15 18:21:00	ENUM{{1 LOADED}}	0	EHS-NMC-18-00098	Fire
2018-01-15 18:22:00	ENUM{{0 UNLOADED}}	1	NA	NoFire
2018-01-15 18:23:00	ENUM{{0 UNLOADED}}	1	NA	NoFire
2018-01-15 18:24:00	ENUM{{0 UNLOADED}}	1	NA	NoFire
2018-01-16 00:30:00	ENUM{{1 LOADED}}	0	EHS-NMC-18-00099	Fire
2018-01-16 00:32:00	ENUM{{0 UNLOADED}}	1	NA	NoFire
2018-01-16 00:33:00	ENUM{{0 UNLOADED}}	1	NA	NoFire
2018-01-16 00:34:00	ENUM{{0 UNLOADED}}	1	NA	NoFire
2018-01-16 00:35:00	ENUM{{0 UNLOADED}}	1	NA	NoFire
2018-01-16 00:36:00	ENUM{{0 UNLOADED}}	1	NA	NoFire
2018-01-16 10:08:00	ENUM{{1 LOADED}}	0	EHS-NMC-18-00113	Fire
2018-01-16 10:10:00	ENUM{{0 UNLOADED}}	1	NA	NoFire
2018-01-16 10:11:00	ENUM{{0 UNLOADED}}	1	NA	NoFire
2018-01-16 10:12:00	ENUM{{0 UNLOADED}}	1	NA	NoFire
2018-01-17 15:31:00	ENUM{{1 LOADED}}	0	EHS-NMC-18-00117	Fire
2018-01-17 15:32:00	ENUM{{0 UNLOADED}}	1	NA	NoFire
2018-01-17 15:33:00	ENUM{{0 UNLOADED}}	1	NA	NoFire
2018-01-17 15:34:00	ENUM{{0 UNLOADED}}	1	NA	NoFire
2018-01-17 15:35:00	ENUM{{0 UNLOADED}}	1	NA	NoFire
2018-01-17 18:26:00	ENUM{{1 LOADED}}	0	EHS-NMC-18-00118	Fire
2018-01-17 18:27:00	ENUM{{0 UNLOADED}}	1	NA	NoFire
2018-01-17 18:28:00	ENUM{{0 UNLOADED}}	1	NA	NoFire
2018-01-18 05:10:00	ENUM{{1 LOADED}}	0	EHS-NMC-18-00122	Fire
2018-01-18 05:11:00	ENUM{{0 UNLOADED}}	1	NA	NoFire
2018-01-18 05:12:00	ENUM{{0 UNLOADED}}	1	NA	NoFire

Table 8: Unaccounted records in Fire Event Mapping

Date_Time	TM2_DE_TopDoctorStatus_ZI_6860A.PV	TM2_DE_DryerDoctorStatus_ZI_6860.PV	Top_Blade_Change_Event	Bottom_Blade_Change_Event	Fire_Event_Id	Event
2018-01-16 06:51:00	ENUM{{1 LOADED}}	ENUM{{1 LOADED}}	0	0	EHS-NMC-18-00114	Fire
2018-01-19 03:21:00	ENUM{{1 LOADED}}	ENUM{{1 LOADED}}	0	0	EHS-NMC-18-00128	Fire
2018-01-23 17:59:00	ENUM{{1 LOADED}}	ENUM{{1 LOADED}}	0	0	EHS-NMC-18-00157	Fire
2018-04-10 23:25:00	ENUM{{1 LOADED}}	ENUM{{1 LOADED}}	0	0	EHS-NMC-18-00507	Fire
2018-05-20 15:22:00	ENUM{{1 LOADED}}	ENUM{{1 LOADED}}	0	0	EHS-NMC-18-00790	Fire
2018-06-19 04:08:00	ENUM{{1 LOADED}}	ENUM{{1 LOADED}}	0	0	EHS-NMC-18-00950	Fire
2018-06-29 12:04:00	ENUM{{1 LOADED}}	ENUM{{1 LOADED}}	0	0	EHS-NMC-18-01023	Fire
2018-11-09 00:04:00	ENUM{{1 LOADED}}	ENUM{{1 LOADED}}	0	0	EHS-NMC-18-01751	Fire
2018-11-09 06:00:00	ENUM{{1 LOADED}}	ENUM{{1 LOADED}}	0	0	EHS-NMC-18-01751b	Fire
2018-12-18 15:00:00	ENUM{{1 LOADED}}	ENUM{{1 LOADED}}	0	0	EHS-NMC-18-02061	Fire
2019-02-16 19:26:00	ENUM{{1 LOADED}}	ENUM{{1 LOADED}}	0	0	EHS-NMC-19-00346	Fire
2019-05-27 12:19:00	ENUM{{1 LOADED}}	ENUM{{1 LOADED}}	0	0	EHS-NMC-19-00832	Fire
2019-06-11 07:10:00	ENUM{{1 LOADED}}	ENUM{{1 LOADED}}	0	0	EHS-NMC-19-00915	Fire
2019-08-07 02:11:00	ENUM{{1 LOADED}}	ENUM{{1 LOADED}}	0	0	EHS-NMC-19-01248	Fire
2019-08-19 03:32:00	ENUM{{1 LOADED}}	ENUM{{1 LOADED}}	0	0	EHS-NMC-19-01305	Fire
2019-12-04 20:14:00	ENUM{{1 LOADED}}	ENUM{{1 LOADED}}	0	0	EHS-NMC-19-01825	Fire

Table 6: Leading - Bottom Blade Change and Fire Event Mapping

Date_Time	TM2_DE_DryerDoctorStatus_ZI_6860.PV	Bottom_Blade_Change_Event	Fire_Event_Id	Event
2018-01-15 18:21:00	ENUM{{1 LOADED}}	0	EHS-NMC-18-00098	Fire
2018-01-15 18:27:00	ENUM{{0 UNLOADED}}	1	NA	NoFire
2018-01-16 00:30:00	ENUM{{1 LOADED}}	0	EHS-NMC-18-00099	Fire
2018-01-16 00:38:00	ENUM{{0 UNLOADED}}	1	NA	NoFire
2018-01-16 10:08:00	ENUM{{1 LOADED}}	0	EHS-NMC-18-00113	Fire
2018-01-16 10:13:00	ENUM{{0 UNLOADED}}	1	NA	NoFire
2018-01-17 15:31:00	ENUM{{1 LOADED}}	0	EHS-NMC-18-00117	Fire
2018-01-17 15:38:00	ENUM{{0 UNLOADED}}	1	NA	NoFire
2018-01-17 18:26:00	ENUM{{1 LOADED}}	0	EHS-NMC-18-00118	Fire
2018-01-17 18:30:00	ENUM{{0 UNLOADED}}	1	NA	NoFire
2018-01-17 18:31:00	ENUM{{0 UNLOADED}}	1	NA	NoFire
2018-01-18 05:10:00	ENUM{{1 LOADED}}	0	EHS-NMC-18-00122	Fire
2018-01-18 05:14:00	ENUM{{0 UNLOADED}}	1	NA	NoFire
2018-01-18 05:15:00	ENUM{{0 UNLOADED}}	1	NA	NoFire
2018-01-19 22:02:00	ENUM{{1 LOADED}}	0	EHS-NMC-18-00136	Fire
2018-01-19 22:06:00	ENUM{{0 UNLOADED}}	1	NA	NoFire
2018-01-19 22:07:00	ENUM{{0 UNLOADED}}	1	NA	NoFire
2018-02-08 23:20:00	ENUM{{1 LOADED}}	0	EHS-NMC-18-00228	Fire
2018-02-08 23:27:00	ENUM{{0 UNLOADED}}	1	NA	NoFire
2018-02-11 03:38:00	ENUM{{1 LOADED}}	0	EHS-NMC-18-00239	Fire
2018-02-11 03:42:00	ENUM{{0 UNLOADED}}	1	NA	NoFire
2018-02-11 03:43:00	ENUM{{0 UNLOADED}}	1	NA	NoFire
2018-02-11 10:47:00	ENUM{{1 LOADED}}	0	EHS-NMC-18-00242	Fire
2018-02-11 10:55:00	ENUM{{0 UNLOADED}}	1	NA	NoFire
2018-02-11 10:56:00	ENUM{{0 UNLOADED}}	1	NA	NoFire

Table 7: Lagging - Bottom Blade Change and Fire Event Mapping

Date_Time	TM2_DE_DryerDoctorStatus_ZI_6860.PV	Bottom_Blade_Change_Event	Fire_Event_Id	Event
2018-01-25 09:59:00	ENUM{{0 UNLOADED}}	1	NA	NoFire
2018-01-25 10:15:00	ENUM{{1 LOADED}}	0	EHS-NMC-18-00161	Fire
2019-08-09 14:31:00	ENUM{{0 UNLOADED}}	1	NA	NoFire
2019-08-09 14:32:00	ENUM{{0 UNLOADED}}	1	EHS-NMC-19-01264	Fire

Other challenges

- Data drift observed in 2018 and 2019 files
 - Mean and variance shift for several variables
- Model building – Several models tried for detecting fire events with a lead time of 10 minutes: LSTM, Markov models etc.
 - Failed to get an accuracy above 45%-50%.
 - Sensitivity of the model below 45%

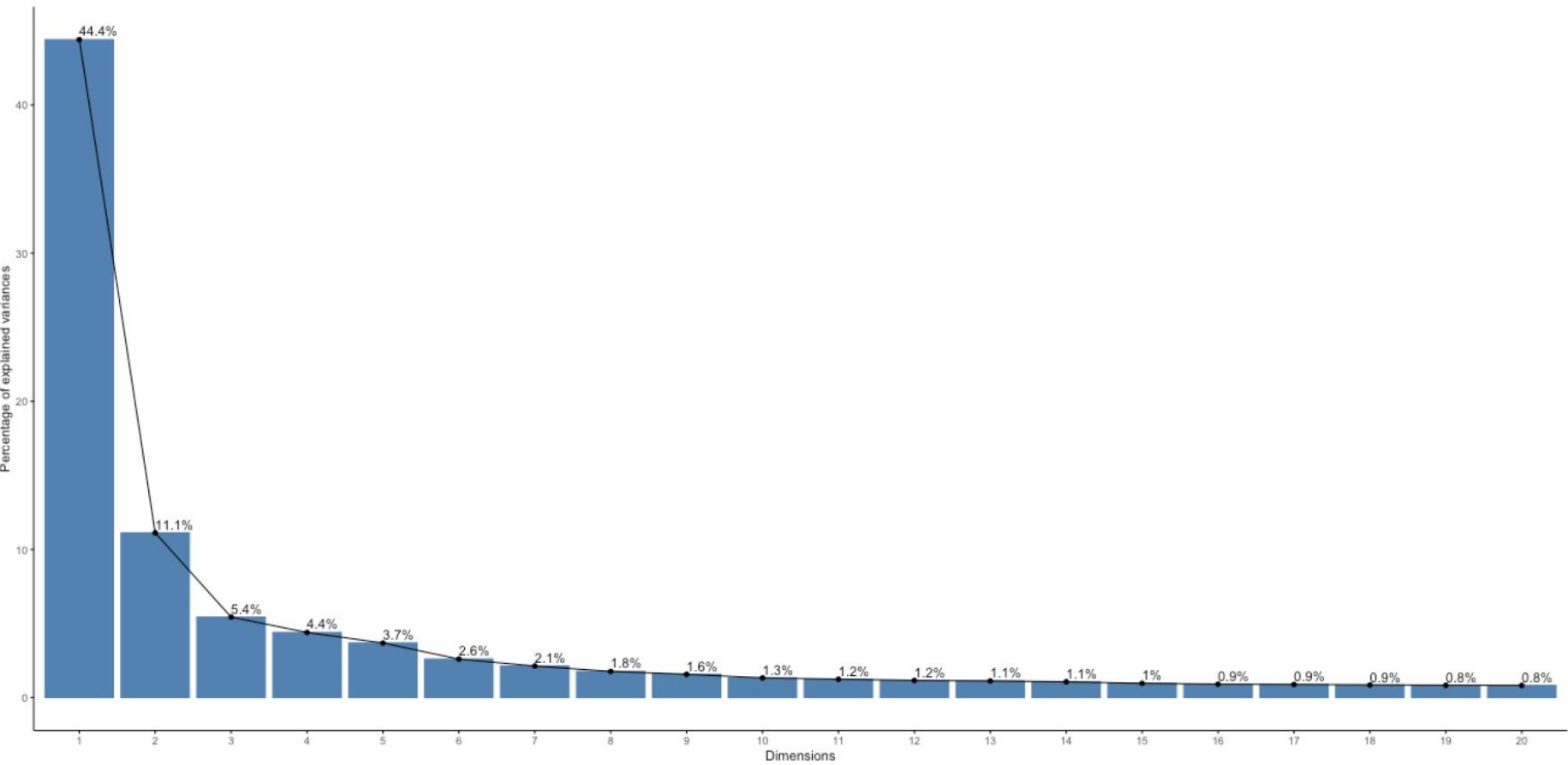
Data Exploration using PCA

- Can we really predict the event with 134 variables?
 - Curse of dimensionality – data sparsity, generalization, high variance.
 - Only 60 events of interest out of approx. 600,000 observations spanning 2 years.
 - Mean and variance shift in the data between 2018 and 2019.
- Eigenvalues can be used to determine the number of principal components to retain:
 - An eigenvalue > 1 indicates that PCs account for more variance than accounted by one of the original variables in standardized data. This can be used as cut-off to retain the #PCs. This holds true only when the data are standardized.
- Another criteria could be to look into the cumulative variance explained by the PCs.
 - If 70% of the total variance explained is reasonable for business then we can restrict to 6 PCs (in this case).

Table 5: Eigen Value using PCA

	eigenvalue	variance.percent	cumulative.variance.percent
Dim.1	52.8498	44.4116	44.4116
Dim.2	13.2353	11.1221	55.5337
Dim.3	6.4677	5.4351	60.9688
Dim.4	5.2340	4.3983	65.3671
Dim.5	4.3863	3.6860	69.0531
Dim.6	3.0926	2.5988	71.6519
Dim.7	2.5365	2.1315	73.7834
Dim.8	2.1034	1.7676	75.5510
Dim.9	1.8738	1.5746	77.1256
Dim.10	1.5833	1.3305	78.4561
Dim.11	1.4759	1.2402	79.6963
Dim.12	1.3790	1.1589	80.8552
Dim.13	1.3367	1.1233	81.9785
Dim.14	1.2718	1.0687	83.0472
Dim.15	1.1475	0.9643	84.0115
Dim.16	1.0802	0.9077	84.9192
Dim.17	1.0573	0.8884	85.8077
Dim.18	1.0133	0.8515	86.6592
Dim.19	0.9912	0.8329	87.4921
Dim.20	0.9840	0.8269	88.3190
Dim.21	0.9575	0.8046	89.1236
Dim.22	0.9268	0.7788	89.9024
Dim.23	0.9184	0.7718	90.6742
Dim.24	0.8726	0.7333	91.4075
Dim.25	0.8125	0.6828	92.0903
Dim.26	0.8043	0.6759	92.7662
Dim.27	0.7431	0.6245	93.3906
Dim.28	0.7016	0.5895	93.9802
Dim.29	0.6729	0.5655	94.5456
Dim.30	0.6351	0.5337	95.0793

Scree Plot



Other plots of PCA

```
## Principal Component Analysis Results for variables
## =====
##   Name      Description
## 1 "$coord"   "Coordinates for the variables"
## 2 "$cor"     "Correlations between variables and dimensions"
## 3 "$cos2"    "Cos2 for the variables"
## 4 "$contrib" "contributions of the variables"
```

Table 6: Correlation between scaled train data and PCs

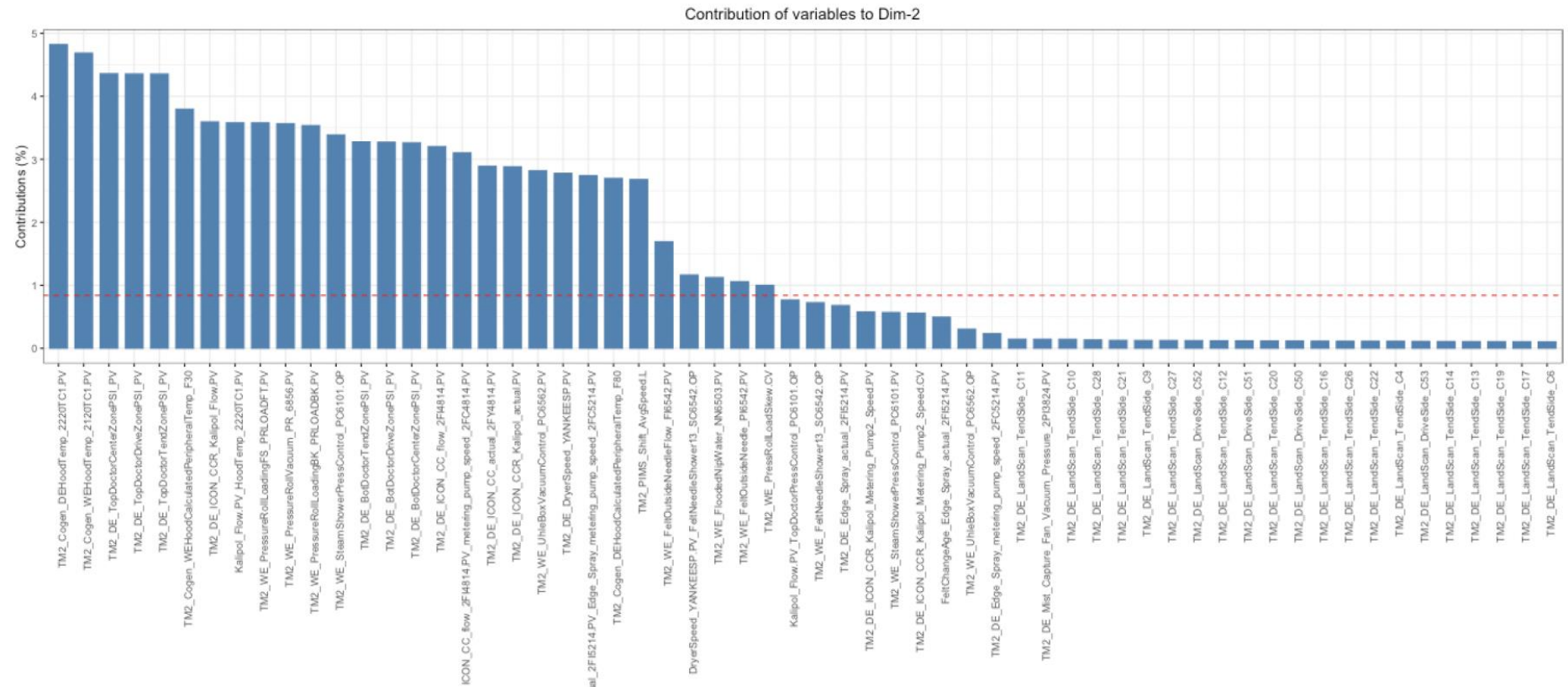
	Dim.1	Dim.2	Dim.3	Dim.4	Dim.5	Dim.6	Dim.7	Dim.8	Dim.9	Dim.10
TM2_Cogen_WEHoodCalculatedPeripheralTemp_F30	0.3438	0.7088	-0.1409	0.0453	-0.0192	-0.1245	0.0886	0.0020	0.0096	-0.0940
TM2_DE_Day1HayoutTime_2HAYOUT1.PV	-0.0638	0.0036	-0.0311	-0.0777	-0.0011	-0.0233	0.0995	0.1058	0.1032	-0.0147
TM2_DE_ICON_CCR_Kalipol_Metering_Pump2_Speed.PV	-0.1180	-0.2764	-0.3683	-0.2047	0.3275	-0.1032	0.4615	0.1262	-0.1178	0.3579
TM2_DE_ICON_CCR_Kalipol_actual.PV	0.2314	0.6177	0.1032	0.1088	-0.3987	0.1594	-0.1272	-0.0895	-0.0545	0.0880
TM2_DE_ICON_CCR_Kalipol_Flow.PV	0.3148	0.6897	-0.0416	-0.0094	-0.4008	0.1615	-0.1582	0.0374	-0.1616	0.2008

Table 8: Variable contribution to PCs

	Dim.1	Dim.2	Dim.3	Dim.4	Dim.5	Dim.6	Dim.7	Dim.8	Dim.9	Dim.10
TM2_Cogen_WEHoodCalculatedPeripheralTemp_F30	0.2236	3.7956	0.3070	0.0392	0.0084	0.5013	0.3097	0.0002	0.0049	0.5586
TM2_DE_Day1HayoutTime_2HAYOUT1.PV	0.0077	0.0001	0.0150	0.1155	0.0000	0.0175	0.3902	0.5320	0.5684	0.0136
TM2_DE_ICON_CCR_Kalipol_Metering_Pump2_Speed.PV	0.0263	0.5772	2.0974	0.8003	2.4449	0.3444	8.3982	0.7568	0.7407	8.0904
TM2_DE_ICON_CCR_Kalipol_actual.PV	0.1013	2.8824	0.1645	0.2263	3.6233	0.8220	0.6376	0.3806	0.1584	0.4896
TM2_DE_ICON_CCR_Kalipol_Flow.PV	0.1875	3.5936	0.0268	0.0017	3.6617	0.8437	0.9861	0.0665	1.3938	2.5470

Contains the contributions (in percentage) of the variables to the principal components - $(\text{var.cos2} * 100) / (\text{total cos2 of the component})$

- Interpretation of plot: The red line denotes the expected average contribution assuming each variable contributes uniformly to the PC.
 - In this case, the expected average contribution will be $1/119 = 0.8403361\%$.
- The below graphs visualize the top 60 variables contributing to second PCs.



Variable contribution to multiple PCs

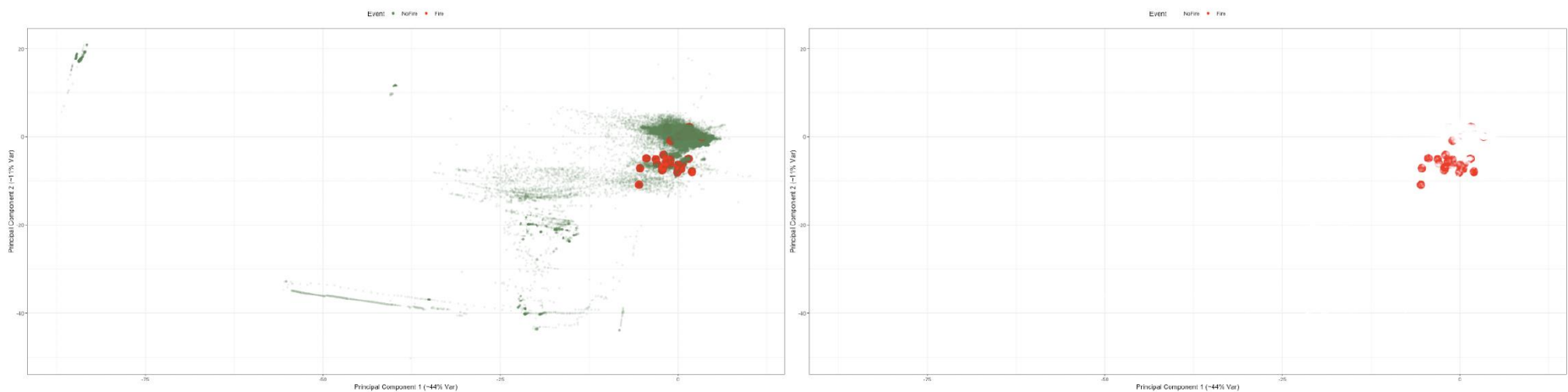
- When two components are taken together and the top variables contributing to the two components are to be computed, the computation goes like this
 - $\text{contrib} = [(C1 * \text{Eig1}) + (C2 * \text{Eig2})]/(\text{Eig1} + \text{Eig2})$
C1 and C2 are the contributions (var\$contrib) of the variable on PC1 and PC2, respectively Eig1 and Eig2 are the eigenvalues of PC1 and PC2, respectively.

Table 9: Total contribution of Variables to PC1 and PC2

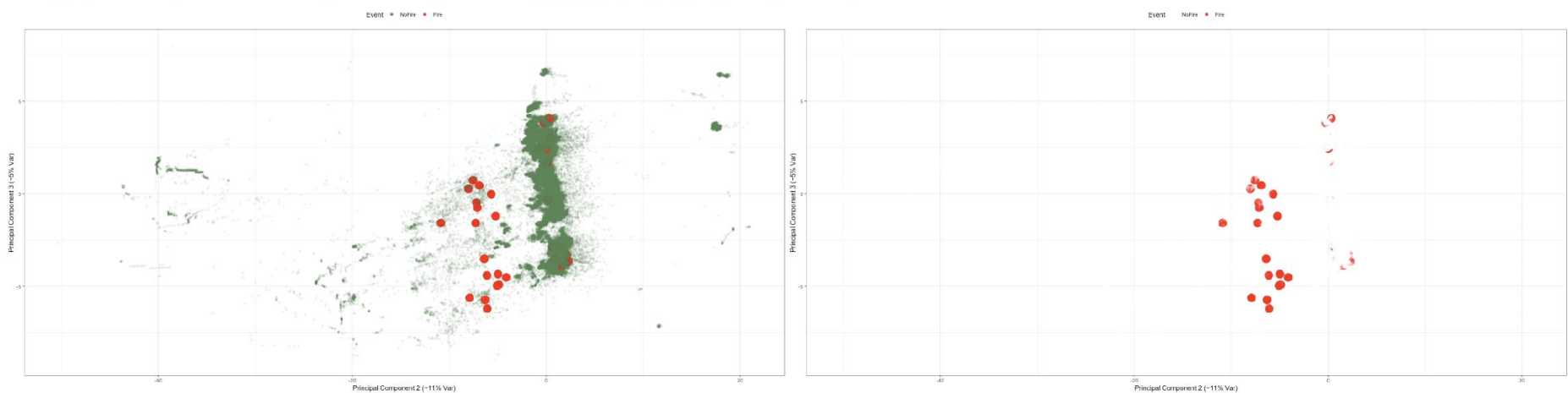
name		contrib
TM2_Cogen_WEHoodCalculatedPeripheralTemp_F30	TM2_Cogen_WEHoodCalculatedPeripheralTemp_F30	0.9390
TM2_DE_Day1HayoutTime_2HAYOUT1.PV	TM2_DE_Day1HayoutTime_2HAYOUT1.PV	0.0062
TM2_DE_ICON_CCR_Kalipol_Metering_Pump2_Speed.PV	TM2_DE_ICON_CCR_Kalipol_Metering_Pump2_Speed.PV	0.1367
TM2_DE_ICON_CCR_Kalipol_actual.PV	TM2_DE_ICON_CCR_Kalipol_actual.PV	0.6583
TM2_DE_ICON_CCR_Kalipol_Flow.PV	TM2_DE_ICON_CCR_Kalipol_Flow.PV	0.8697

Individual observation plot – Why model error will be high?

The plot of the individual observation on the principal components (1 and 2):

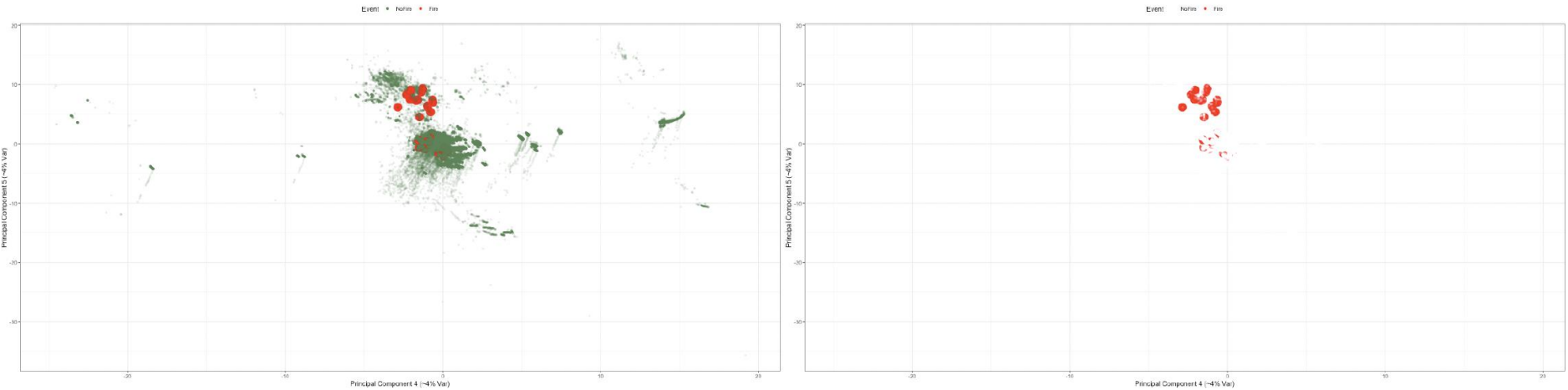


The plot of the individual observation on the two principal components (2 and 3):

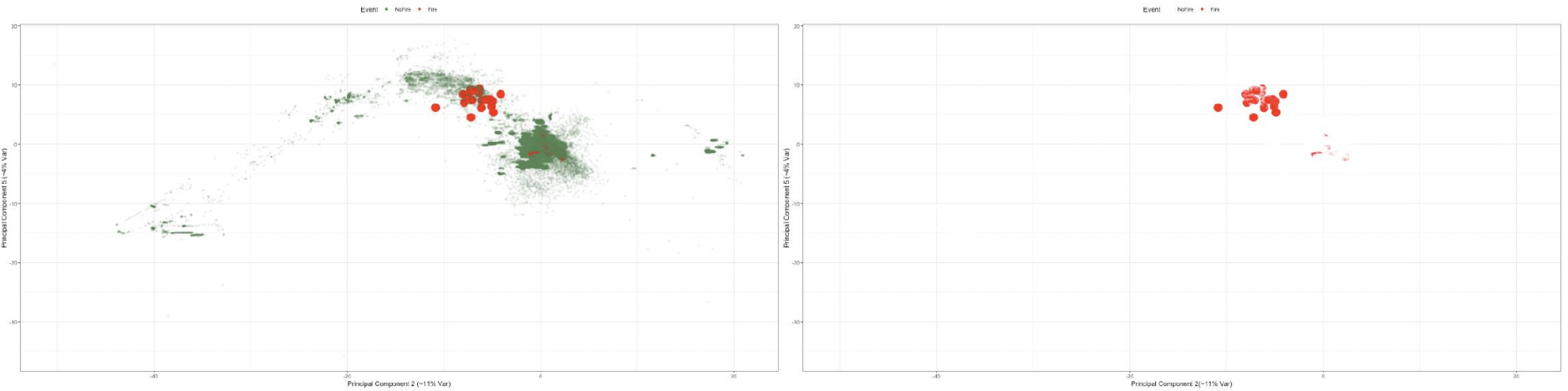


Individual observation plot – Why model error will be high?

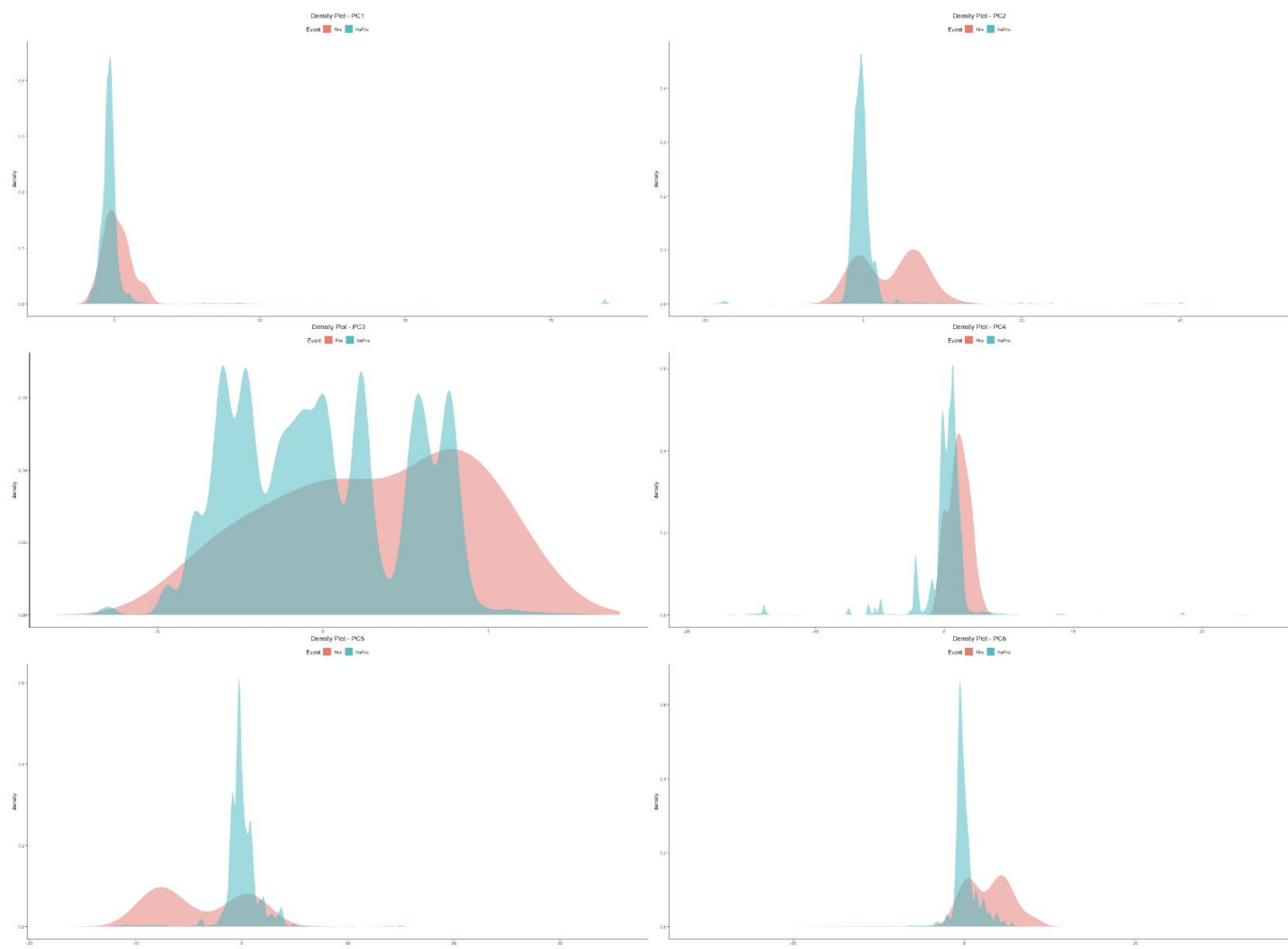
The plot of the individual observation on the two principal components (4 and 5):



The plot of the individual observation on the two principal components (2 and 5):



Distribution plot of PC1 to PC6

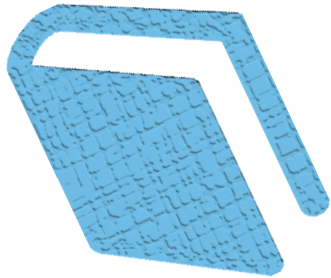


Conclusion

- Reduced the feature space for 134 dimensions to less than 25 explaining 80% of the variation in the data.
- Visuals help explain – why we can have a model with better accuracy.
- Used the first 20 PCs to build the LSTM model
 - Sensitivity of approx. 75% achieved.
- Built a second order Markov chain using the 2 PCs components.
 - PC2 and PC5 used to calculate the joint prob distribution.
 - Used Gaussian model to understand the threshold at which event of fire was marked as “Low”, “High” using the joint prob dist.
 - Build TFM for months Jan to Aug using second order Markov chain
 - Test of Independence and test of homogeneity for Markov property.
 - Null Hypothesis: The sequence of transitions are independent Alternate Hypothesis: The sequence of transitions are dependent
 - Sensitivity of 100 percent on test data (Sep to Dec).
 - Model accuracy: approx. 70%
 - In the long run, the fire event will reach a limiting value of 2.7%.

Summary

Summary of the topics covered in this lesson:



- Unsupervised learning has several applications
- Clustering is one of the most used unsupervised learning algorithm.
- Hierarchical clustering is useful when comparing various brands, products on certain parameters.
- K means clustering is useful when the number of observations runs in thousands say customer footfall into supermarket, bank etc.
- Both Hierarchical and K means clustering with Euclidian distance measure, cannot be used for grouping data with categorical variable.
- Several distance measuring techniques exists.
- PCA has several applications including data exploration, usages in supervised learning, missing data, recommender system.

End of Lesson05–Unsupervised learning

